Deep Integration of Export Destinations and Firm Innovation: Evidence from China

Zhen Yu[1], Chenxi Li[2], Xue Li[3], Xuan Luo[4]

[1] Institute of Economy of USA and Canada, Wuhan University. Email: Yuzhenwhu@126.com

[2] School of Economics and Management, Tsinghua University. Email: licx@sem.tsinghua.edu.cn

[3] Institute of Economy of USA and Canada, Wuhan University. Email: lixue071123@163.com

[4] School of Economics, University of Nottingham Ningbo China. Email: luoxuan.econ@hotmail.com. Corresponding author.

Deep Integration of Export Destinations and Firm Innovation: Evidence from China

Abstract

How do firms in non-member countries adjust their innovation strategies when their export destinations engage in deep integration through membership in preferential trade agreements (PTAs)? Using global PTA provisions and Chinese firm-level data, we construct two shift-share measures of local industries' exposure to destination integration, based on the average depth of PTAs engaged in by export destinations and the number of their PTA partners. Only the PTA depth exposure affects innovation: firms tend to reduce their probability of filing invention patents and reorient their technological direction. Neither measure has a significant effect on patenting counts along the intensive margin. Various negative trade effects along the intensive and extensive margins provide evidence of competition in export destinations, consistent with the Schumpeterian effect. We also find that the most productive firms innovate and profit more, whereas the least productive firms innovate less and incur losses, in response to the deep integration of their export destinations, supporting the escape-competition argument. In addition, we show that firms' innovation responses to the deep integration of export destinations vary by sectoral R&D intensity, PTA provisions, China's PTA partnerships, and patent types, among others. The negative innovation effect, coupled with the deterioration of the terms of trade, suggests potential welfare losses for non-member countries.

Keywords: Innovation, Schumpeterian effect, Escape-competition effect, Preferential trade agreements, Deep integration

JEL codes: F13, F14, F15, F61, O31

1 Introduction

The rapid expansion and deepening of preferential trade agreements (PTAs) have reshaped not only trade patterns but also the dynamics of global economic activity. Extensive literature examines the effects of PTAs on trade creation, innovation, and investment within member countries.[1] However, the literature has primarily focused on the trade diversion effects for non-member countries when their export destinations engage in deep integration through PTA membership (Dai et al., 2014; Mattoo et al., 2022; Lee et al., 2023), with much less attention given to the third-country effects on other related economic variables such as innovation. Beyond trade, third-country firms' profitability, learning scope or access to knowledge could also be affected.[2] This, in turn, may alter their innovation incentives. Alternatively, these firms with different productivity levels may exhibit heterogeneous innovation strategies to navigate the intensified competition in the export markets. Understanding these complexities is essential for assessing the broader economic effects of PTAs.

In this paper, we fill this gap by studying how firms in third countries adjust their innovation behavior when their export destinations undergo deep integration through PTAs. Our analysis adopts China as a third country. Studying the innovation response of Chinese firms is important in its own right, as China is the largest developing economy and the second-largest economy in the world. Although China is a major global exporter, its PTA partners only account for around 6% of world GDP in 2013, and it has limited influence over the rules governing deep integration among other countries. As such, it is difficult for Chinese firms to manipulate the inter-connectedness and deep integration of export destinations. The exposure to the evolution of external destination integration is thus exogenous to Chinese firms, allowing for a credible identification strategy.

We first construct the exposure of local industries in China to integration among their export destinations using the World Bank Deep Trade Agreements Database (Hofmann et al., 2017). We distinguish two margins of destination PTA engagement, which serve as shifts in a shift-share design. The intensive margin captures the average depth of a destination's PTAs with countries other than China, while the extensive margin captures the number of its PTA partners. We then construct two

[1] For example, Baier and Bergstrand (2007), Campi and Dueñas (2019), Baier et al. (2019), and Dutt (2020) study the trade effects of PTAs; Bustos (2011) studies the innovation effect of PTAs; and Baltagi et al. (2008) and Medvedev (2012) study investment effects of PTAs within member countries.

[2] Trade itself can serve as a conduit for learning and imitation. Using Chinese patent citation data, Liang et al. (2024) highlight the learning-by-exporting effect and demonstrate that there are actual knowledge transfers from destination markets to exporting firms, which help firms improve their innovation. Aghion et al. (2023) find that French firms disseminate knowledge to destination countries through exporting.

shift-share measures, the PTA depth exposure and the PTA breadth exposure, by weighting these destination-level shifts using predetermined city–industry export shares to each destination. This allows us to disentangle the depth of PTA integration from the expansion of PTA networks.

To conduct the empirical analysis, we first merge the patent filing data from the State Intellectual Property Office of China (SIPO), with the Annual Survey of Industrial Firms dataset (ASIF) that includes rich firm-level operation information. We focus on invention patent filings by manufacturing firms. Then we proceed to merge the matched data above with the two shift-share measures constructed. The final data allow us to study the innovation effects when Chinese firms respond to the destination integration.

We find that the PTA depth exposure reduces the probability of invention patenting along the extensive margin but has no significant effect on the number of patents filed along the intensive margin. The PTA breadth exposure has no significant effect on either margin of innovation. Therefore, we use the probability of invention patenting as our baseline measure of innovation throughout the paper. To isolate the innovation effect of destination integration, we control for tariff exposure and destination-market-size exposure faced by each local industry, various firm-level characteristics, and a rich set of fixed effects. The negative innovation effect of the PTA depth exposure is comparable in magnitude to that of the tariff exposure. We further conduct a series of diagnostic tests that support the plausibility of our identifying assumption and alleviate concerns that the negative innovation effect of the PTA depth exposure is driven by endogenous shifts, shares, or pre-existing trends. The results are also robust across different subsamples and alternative measures of destination integration.

We explore potential mechanisms through which the deep integration of export destinations affects the innovation behavior of Chinese firms. Both the Schumpeterian effect and the escape-competition effect may be at work. The Schumpeterian effect emphasizes how intensified competition reduces the potential rents that firms can capture from innovation, thereby discouraging innovation (Schumpeter, 1942). To substantiate the competitive pressure faced by Chinese firms in export markets, we examine how the PTA depth exposure affects the intensive and extensive margins of trade. We find that the PTA depth exposure reduces export values primarily through lower prices, consistent with intensified price competition arising from trade diversion. Meanwhile, exporters reduce their product scope, although the number of export destinations does not change significantly. We also find that export prices decline more than import prices, suggesting a deterioration in China's terms of trade.

On the other hand, the escape-competition effect predicts that more productive firms will innovate more to escape competition from their rivals (Aghion et al., 2005). We find the least productive firms suffer profit losses and exhibit the greatest negative innovation response to destination integration. Interestingly, the negative effects are alleviated for more productive firms. And the more productive firms profit more, innovate more. Notably, these firms nevertheless experience declines in profit rates and markups, providing further evidence of intensified competitive pressure despite their stronger innovation responses.

We further conduct several heterogeneity analyses to shed light on the underlying mechanisms. First, we examine heterogeneous effects by PTA provisions. Preferential provisions reduce innovation primarily among low-productivity firms, consistent with trade diversion. By contrast, positive spillovers from non-discriminatory provisions stimulate innovation among highly productive firms, which have stronger incentives to innovate to escape competition. We also find that the escape-competition effect is more pronounced in response to the deep integration of China's PTA partners and in R&D-intensive industries. Turning to patent types, we find no statistically significant effects on utility or design patenting. Regarding ownership, the negative innovation effect is statistically significant only for private firms, but not for state-owned or foreign-owned firms. We further find that the negative innovation response is more pronounced in upstream industries, which are more exposed to trade diversion in intermediate-input markets. Finally, the innovation effects of deep integration among export destinations critically depend on the strength of local intellectual property rights enforcement.

Finally, we use firms' four-digit International Patent Classification (IPC4) portfolios to examine technological direction. Deeper integration reduces entry into new IPC classes and the distance from the initial technology portfolio on average. However, firms serving multiple export destinations exhibit a more positive response than single-destination exporters, consistent with their greater scope to adjust innovation across markets with heterogeneous preferences or product standards. Taken together, the results suggest that the deep integration of export destinations not only shifts overall demand and intensifies competitive pressure faced by Chinese firms but also shapes the technological direction of their innovation.

This paper mainly contributes to two strands of the literature. Our first contribution is to the literature on the effects of trade agreements on non-members. Since Viner (1950), this literature has studied trade creation within agreements and trade diversion away from third countries (Dai

et al., 2014; Conconi et al., 2018; Maskus and Ridley, 2021; Mattoo et al., 2022; Lee et al., 2023; He and Luo, 2026). Theory also links trade diversion to a deterioration in a third country's terms of trade and welfare (Panagariya, 2000; Mossay and Tabuchi, 2015). Only a few papers study the third-country effects on other economic activities. Based on an incomplete contract model with endogenous contracting institutions, Luo and Hsu (2024) find a negative third-country effect on institutional quality when member countries trade liberalize. We show that PTAs also affect the propensity to file invention patents and the composition of technological portfolios in a non-member economy, and examine heterogeneous innovation responses along various dimensions.

Second, this paper contributes to the rich literature on the innovation effects of trade. Export market opportunities motivate firms to innovate more (Lileeva and Trefler, 2010; Bustos, 2011; Aw et al., 2011; Liu and Ma, 2020; Liang et al., 2024). The effects of market size and competition on innovation are heterogeneous, depending on firms' productivity (Aghion et al., 2024) and their positions along the quality ladder (Lim et al., 2022). See also Shu and Steinwender (2019) and Melitz and Redding (2021) for surveys of the innovation effects of trade. We differ from this literature by focusing on the inter-connectedness of export destinations through PTAs, rather than changes in their own markets. While both can operate as demand shifters and affect the scale of innovation, deep integration of export destinations contains additional dimensions that a single demand shifter does not capture. By separating PTA depth from the breadth of destination PTA networks and further distinguishing preferential from non-discriminatory provisions, we uncover how different dimensions of destination integration affect Chinese firms through distinct patterns of trade and innovation adjustment. More importantly, we show that destination integration not only affects the scale of innovation but also shapes its technological direction, potentially reflecting firms' responses to heterogeneous preferences or product standards across export markets.

The rest of the paper is structured as follows. Section 2 introduces the data and the construction of local industries' exposure to the deep integration of export destinations. Section 3 describes the empirical strategy and presents the estimation results, together with identifying assumption checks and robustness checks. Section 4 explores potential mechanisms. Section 5 further examines heterogeneous effects along various dimensions to shed light on the underlying mechanisms and Section 6 investigates how destination integration affects the technological direction of innovation. Section 7 concludes.

2 Data and descriptive statistics

The empirical analysis is mainly based on four datasets. We first describe the datasets, and proceed to construct the measure of the deep integration of export destinations. Lastly, we explain the matching strategy and present the summary statistics.

2.1 Data

We obtain the patent filing data from the State Intellectual Property Office of China (SIPO). For each filing since 1985, the data report the filing date, patent type and title, citations, applicant names and addresses, publication number, International Patent Classification (IPC) codes and the primary IPC code, among others.

China recognizes three types of innovations—invention, utility, and design—with each governed by different requirements under Chinese Patent Law. Invention patents undergo a rigorous examination process that assesses their utility, novelty, and non-obviousness. The innovation must exhibit "prominently substantive characteristics and significant improvement" compared to existing technologies in order to be granted. In contrast, utility model and design patents are typically granted on a registration basis. Design patents primarily address a product's appearance or shape, while utility patents still have to pass a novelty test and meet standards for practicality and functionality. We focus on invention patents in this paper.

The second data source is the Annual Survey of Industrial Firms (ASIF) dataset from 2000 to 2013, as maintained by the National Bureau of Statistics of China. It is the most comprehensive firm-level dataset in China, including all firms with annual sales above 5 million RMB (around US $600,000 at the 2002 exchange rate). The threshold increased to 20 million RMB in 2011. In addition, it includes all state-owned firms with sales below 5 million RMB before 2007. ASIF contains the basic information for each surveyed firm, such as its identification number, location code, and industry affiliation, together with financial and operational information extracted from accounting statements, such as sales, employment, materials, fixed assets, age, capital, and ownership, among others. We only focus on manufacturing firms in this paper.

Thirdly, we use the World Bank Deep Trade Agreements Database to capture the depth of trade agreements between country pairs (Hofmann et al., 2017). It maps the coverage of 52 selected policy

areas in 279 PTAs notified to the WTO and signed between 1958 and 2015. These policy areas are further divided into two groups of 14 WTO+ and 38 WTO-X areas. WTO+ provisions in PTAs reconfirm existing commitments and, in some cases, provide for additional obligations. WTO-X provisions refer to policy areas that are not yet regulated by the WTO. As many of the 52 provisions go beyond trade issues, we focus on 18 core provisions defined by Hofmann et al. (2017): all WTO+ provisions and 4 WTO-X provisions (competition policy, investment, movement of capital and intellectual property rights). We will use these provisions to construct the local exposure of an industry to the deep integration of its export destinations in the next subsection.

The fourth dataset is the Chinese Customs Data, which are monthly reports covering the universe of Chinese export activities from 2000 to 2016. The key variables include export and import values, quantities, firm location, Harmonized System (HS) code at the 8-digit level, destination and firm name, among others. We aggregate the monthly transaction data to firm-destination-product-year level to avoid seasonality and irregular turnover patterns.[3]

We complement these datasets with other datasets, including tariffs compiled by Teti (2024), the Penn World Table, the Worldwide Governance Indicators (WGI), among others, to construct variables of interest and controls, which we will explain shortly.

2.2 Exposure to destination integration at the city-industry level

There has been a notable surge in the signing of PTAs across the globe. The number of PTAs grew to 279 in 2015 from 20 in 1990. Meanwhile, newer agreements are "deeper" in the sense that they generally cover a wider set of policy areas than older agreements (Mattoo et al., 2022). These trends can change not only trade flows among member countries and third countries, but also other related economic activities through trade.

In this paper, we study the third-country effect of the deep integration of export destinations on innovation, with China as the third country. Specifically, we construct the exposure to the deep integration of export destinations confronting local industries in China, and study how it affects China's innovation at the firm level.

Following Hofmann et al. (2017), we define the PTA depth as the number of core provisions

[3]Product is defined at the HS 6 digit level, the most disaggregated codes that are consistently available for our entire study period.

covered in an agreement, and all country pairs within the same PTA have the same depth. More specifically, the depth of a PTA v is $\text{CoreDepth}_t^v = \sum_{i=1}^{18} \text{provision}_{it}^v$, where provision_{it}^v takes the value 1 if core provision i is covered in PTA v at year t, and 0 otherwise.[4] The PTA depth for country pair c and n is thus $\text{CoreDepth}_{cnt} = \text{CoreDepth}_t^v$, where country c and n are both in the trade agreement v. We also consider the depth of a PTA based on the provision's legal enforceability or the sum of all 52 provisions in the robustness checks.

To disentangle the depth of preferential trade agreements (PTAs) signed by China's export destinations (the intensive margin) from the extent of their PTA networks (the extensive margin), we construct two measures of destination-level PTA engagement for China's export destination c. Specifically,

$$\text{DepthEngage}_{ct} = \begin{cases} \dfrac{\sum\limits_{n \neq c,\text{China}} \text{CoreDepth}_{cnt}}{\sum\limits_{n \neq c,\text{China}} \mathbb{I}(\text{PTA}_{cnt} = 1)}, & \text{if } \sum\limits_{n \neq c,\text{China}} \mathbb{I}(\text{PTA}_{cnt} = 1) > 0, \\ 0, & \text{if } \sum\limits_{n \neq c,\text{China}} \mathbb{I}(\text{PTA}_{cnt} = 1) = 0, \end{cases}$$

and

$$\text{BreadthEngage}_{ct} = \sum_{n \neq c,\text{China}} \mathbb{I}(\text{PTA}_{cnt} = 1).$$

where n indexes countries other than China and destination c, and $\mathbb{I}(\text{PTA}_{cnt} = 1)$ indicates that c and n are members of a PTA in year t. Here, DepthEngage_{ct} measures the average depth of the PTAs in which export destination c engages, while $\text{BreadthEngage}_{ct}$ captures the number of its PTA partners, excluding China. As a robustness check, we also calculate a trade-weighted average depth, using initial trade flows with each partner as weights.

The exposure to PTA engagement of export destinations could exert differential impacts on local economic activities owing to the local comparative advantages of different industries. Following Bartik (1991), we proceed to construct two shift-share measures of destination integration (DI) confronting a city-industry pair as follows:

$$DI_{rjt}^{depth} = \sum_c Share_{rjc,0} \text{DepthEngage}_{ct}, \tag{1}$$

[4] Hofmann et al. (2017) also assign values to provision_{it}^v based on the provision's legal enforceability: 0 if the provision is not mentioned; 1 if the provision is mentioned, legally enforceable but explicitly excluded by dispute settlement; 2 if the provision is mentioned and legally enforceable.

$$DI_{rjt}^{breadth} = \sum_{c} Share_{rjc,0} \text{BreadthEngage}_{ct}, \tag{2}$$

where r and j represent a city and a Chinese Industry Classification (CIC) 4-digit industry, respectively, and $Share_{rjc,0} = \frac{export_{rjc,2000}}{\sum_c export_{rjc,2000}}$ denotes the share of r's exports to country c relative to its total exports in industry j to the world in the base year, such that $\sum_c Share_{rjc,0} = 1$. We fix the export share in the base year 2000 to reduce endogeneity issues (Autor et al., 2013; Borusyak et al., 2022). Therefore, DI^{depth} is the PTA depth exposure and represents the average deep integration of export destinations along the intensive margin, whereas $DI^{breadth}$ is the PTA breadth exposure and represents the average number of PTA partners across export destinations along the extensive margin.

We merge the DI dataset with the patent filing and ASIF datasets. We relegate the data matching procedure to Appendix A1. Figure 1 shows the distributions of DI^{depth} and $DI^{breadth}$. Table A1 reports the summary statistics.

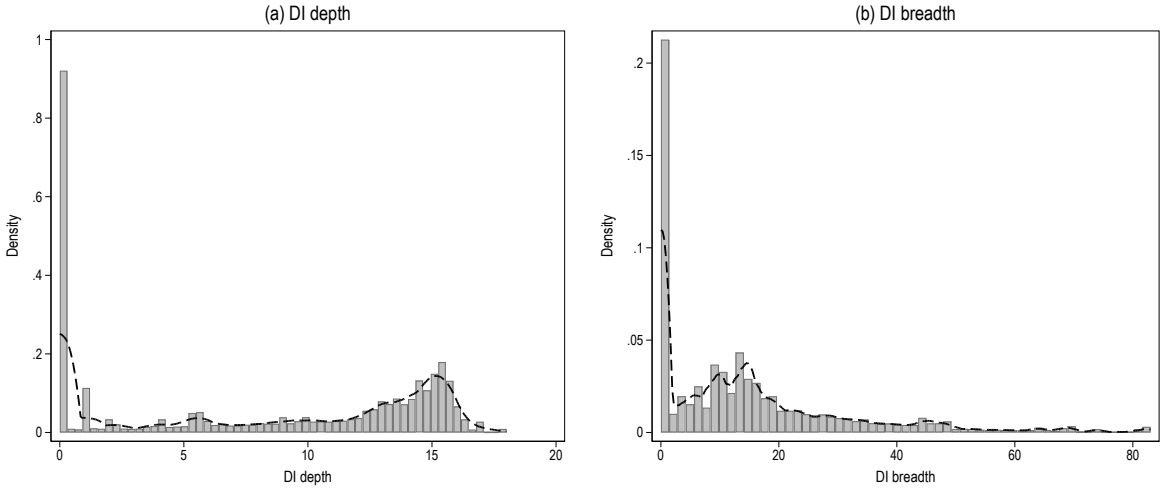


Figure 1: Distributions of DI^{depth} and $DI^{breadth}$ at the city–industry–year level. Panel (a) plots exposure to the average depth of destination-country PTA engagement, and Panel (b) plots exposure to the number of destination-country PTA partners, excluding China.

3 Firm-level analysis

3.1 Econometric specification

Our baseline firm-level econometric specification is given by

$$Y_{f,rjt} = \alpha_0 + \alpha_1 DI_{rjt}^{depth} + \alpha_2 DI_{rjt}^{breadth} + \alpha_3 tariff_{rjt} + FE\ and\ Controls + \varepsilon_{f,rjt}. \tag{3}$$

Here, $Y_{f,rjt}$ denotes either the log number of invention patent filings (the intensive margin) or an indicator for whether the firm files any invention patents (the extensive margin), for firm f located in city r and operating in CIC 4-digit industry j in year t. The variable DI_{rjt}^{depth} measures the exposure of local industry j to the average depth of PTA engagement among export destinations, while $DI_{rjt}^{breadth}$ captures the corresponding exposure to the breadth of their PTA networks, measured by the number of PTA partners (excluding China).

Changes in tariffs over time, including those associated with WTO accession, may be correlated with preferential trade liberalization under PTAs. Following Felbermayr and Teti (2023), we control for the tariffs faced by local industries to distinguish the effects of destination-country PTA engagement from those of changes in tariff barriers. Specifically, we construct $tariff_{rjt} = \sum_c Share_{rjc0} Tariff_{cjt}$, where $Tariff_{cjt}$ denotes the average tariff imposed by destination country c on products in industry j in year t, calculated as the simple average across HS96 products mapped to that industry, using data from Teti (2024). The term $\varepsilon_{f,rjt}$ is the error term.

Our specification includes firm and year fixed effects. Firm fixed effects absorb all time-invariant firm characteristics, so identification relies on within-firm variation over time. Year fixed effects account for common macroeconomic shocks affecting all firms. In addition, *FE and Controls* includes city-year, industry-year, and city-industry fixed effects, as well as other firm-level variables, including age, employment, the capital-labor ratio, and ownership type. To mitigate concerns about bad controls and self-selection, we interact the initial values of these characteristics with year dummies, allowing firms with different initial characteristics to follow different innovation trajectories.

An important distinction is that our main explanatory variables capture the PTA engagement of export destinations with their partners, rather than the economic characteristics of the destinations themselves. In particular, local industries with greater initial exposure to larger export markets

may follow different innovation trajectories, reflecting differences in market opportunities, economies of scale, and learning potential. Moreover, such initial market exposure may be correlated with subsequent changes in destination-country PTA engagement. To account for these potentially confounding factors, we construct the initial export-share-weighted destination-market size as follows: Market Size$_{rj0} = \sum_c Share_{rjc0} GDP_{c0}$, where GDP_{c0} denotes the initial GDP of export destination c, and $Share_{rjc0}$ is the initial share of destination c in exports from industry j in city r. This measure captures the average size of export destination markets to which a local industry was initially exposed, weighted by its initial export distribution across destinations. We interact this measure with year fixed effects to flexibly account for differential innovation trajectories associated with initial exposure to export-market opportunities. To deal with the potential heteroskedasticity and serial autocorrelation, we cluster the standard errors at the city-industry level.

Our identifying assumption is that DI_{rjt}^{depth} and $DI_{rjt}^{breadth}$, conditional on the fixed effects and controls in equation (3), are exogenous to firms' innovation. The reasons underlying the identifying assumption are as follows. Although China is a major global exporter, its PTA partners only account for around 6% of world GDP in 2013. As such, China has limited influence over the rules governing the deep integration among other countries. Hence, it is difficult for Chinese firms to manipulate the inter-connectedness and deep integration of export destinations, allowing for a credible identification strategy.

3.2 Baseline results

Table 1 reports the effects on the extensive and intensive margins of invention patenting. Columns (1)–(4) use an indicator for whether a firm files at least one invention patent in a given year, while column (5) examines firms' first entry into invention patenting. Columns (6)–(9) examine the intensive margin, using the logarithm of the number of invention patent filings conditional on positive filings. Column (10) considers the logarithm of citation-weighted patenting. For each margin, we begin with specifications that control for external market size, city-year fixed effects, industry-year fixed effects, and city-industry fixed effects. We then progressively introduce firm fixed effects and firm characteristics interacted with year dummies. Columns (4) and (9) exclude DI^{depth} to examine the effect of PTA breadth exposure, $DI^{breadth}$, in isolation.

The results show that DI^{depth} significantly reduces the probability of filing an invention patent.

In our preferred extensive-margin specification in column (3), the estimated coefficient is -0.000213, statistically significant at the 1% level. Column (5) further shows that the PTA depth exposure reduces the likelihood of firms entering invention patenting for the first time. By contrast, the coefficients on DI^{depth} in the intensive-margin specifications in columns (6)–(8) are negative but statistically insignificant. Column (10) further examines whether the PTA depth exposure affects citation-weighted innovation output. Following Kogan et al. (2017), we assign each invention patent p a weight of $1 + C_p/\overline{C}_{a(p),k(p)}$, where C_p denotes the cumulative number of forward citations received by patent p, and $\overline{C}_{a(p),k(p)}$ is the average citation count of patents filed in the same application year and belonging to the same primary four-digit International Patent Classification (IPC4) subclass. The estimated coefficient on DI^{depth} remains negative but is statistically insignificant.

The coefficients on the PTA breadth exposure $DI^{breadth}$ are consistently small and statistically insignificant in all columns, whether the breadth exposure is included jointly with the depth exposure or enters the specification alone, as in columns (4) and (9). Table A8 reports Poisson regressions for the invention-patent count, including zeros. The depth and breadth coefficients are statistically insignificant in all three specifications. Taken together, these findings suggest that the negative innovation response operates primarily through the extensive margin of invention patenting and is associated with the depth of PTA integration among export destinations and their partners, rather than the breadth of their PTA networks. We therefore adopt the specification in column (3) as our baseline for the subsequent analyses.

To assess the economic magnitude of the effect, we benchmark the extensive-margin estimate in column (3) against that of tariff exposure. A one-standard-deviation increase in DI^{depth} reduces the probability of invention patenting by 0.084 percentage points, comparable to the 0.071-percentage-point reduction associated with a one-standard-deviation increase in tariff exposure. Although the average effect appears modest, it masks substantial heterogeneity across firms with different productivity levels. As we shall see shortly in Figure 2, the patenting probability increases by 0.3 percentage points for the most productive firms (top TFP quintile), equivalent to 10% of the sample mean patenting probability, whereas it declines by 0.22 percentage points for the least productive firms (bottom TFP quintile), equivalent to 6.7% of the mean.

Table 1: Baseline results

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Extensive margin					Intensive margin				
	Patenting Probability				First entry	log(# patenting)				Log citation-weighted patents
DI^{depth}	-0.000342***	-0.000243***	-0.000213***		-0.000118**	-0.004113	-0.004294	-0.004956		-0.006027
	(0.000090)	(0.000078)	(0.000077)		(0.000052)	(0.002789)	(0.003674)	(0.003654)		(0.004238)
$DI^{breadth}$	-0.000050	0.000014	0.000017	0.000002	-0.000033	0.000695	0.000020	0.000109	-0.000222	0.000081
	(0.000037)	(0.000036)	(0.000036)	(0.000035)	(0.000024)	(0.000695)	(0.000847)	(0.000854)	(0.000826)	(0.000992)
Tariff	-0.031483***	-0.012481*	-0.012173*	-0.008988	-0.017376***	-0.250306	0.596128	0.601476	0.757035*	0.378340
	(0.007609)	(0.006908)	(0.006865)	(0.006782)	(0.004815)	(0.286251)	(0.435047)	(0.433668)	(0.413617)	(0.497035)
Firm FE	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Firm characteristics × year	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
City-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City-Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market size exposure	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,744,560	2,541,619	2,541,619	2,541,619	2,417,568	86,299	65,226	65,222	65,222	65,222
Adjusted R^2	0.105	0.323	0.326	0.326	0.097	0.182	0.412	0.413	0.413	0.331

Notes: Columns (1)–(4) use an indicator for positive invention-patent filings. Column (5) uses first invention-patent entry. Columns (6)–(9) use ln(invention patent count) conditional on positive counts. Columns (4) and (9) omit DI^{depth} and retain $DI^{breadth}$. Column (10) uses the logarithm of the citation-weighted patent count, following Kogan et al. (2017). Standard errors clustered at the initial city–industry level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.3 Identifying assumption checks

To assess the plausibility of the identifying assumption underlying our shift-share design, we conduct a series of diagnostic tests and robustness checks, reported in Appendix A2. First, following Borusyak et al. (2025), we examine potential endogeneity at both the shift and share levels. The negative effect of DI^{depth} remains robust after controlling for potential confounding factors at both levels. Second, we implement exposure-robust inference following Borusyak et al. (2022), and the regression results remain significant at the shift level under alternative clustering schemes. Third, the dynamic analysis reveals no significant pre-trends, while a permutation-based placebo test shows that the observed negative effect is unlikely to arise under random reassignment of PTA histories. Finally, the results remain robust to alternative base-year exposure shares and a recentered IV approach that accounts for nonrandom exposure to the PTA network. Taken together, these findings support the plausibility of our identifying assumption and alleviate concerns that the negative innovation effect of PTA depth exposure is driven by endogenous shocks, initial exposure patterns, or pretrends.

3.4 Robustness checks

In this subsection, we examine whether the baseline finding is robust to different subsamples and alternative constructions of destination PTA engagement. Column (1) of Table 2 repeats the baseline estimate. Columns (2) and (3) restrict the sample to years before the ASIF threshold change and to years after China's WTO accession, respectively. Columns (4)–(9) use alternative measures of the deep integration of export destinations. These results support our baseline finding that the innovation response is driven by the intensive margin of destination integration, rather than by the number of destination-country PTA partners.

Threshold effect — Note that the ASIF dataset includes firms with a designated threshold, which increases from 5 million RMB to 20 million RMB in 2011. To rule out the possibility of our results being affected by the change in the firm size threshold, we focus on the subsample before year 2011, and re-estimate the effect of the depth exposure DI_{rjt}^{depth}. The estimated coefficient is smaller but statistically significant, as shown in column (2) of Table 2.

WTO accession effect — The sample period covers the years 2000 to 2013, and thus the estimation results may capture the effect of China's accession to the WTO in December 2001. To

Table 2: Robustness checks: Alternative subsamples and definitions of depth exposure

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Patent filing dummy								
	Baseline	Year $\leq$ 2010	Year $\geq$ 2002	Weighted	Legal	WTO-plus	Core-extra	Total depth	DESTA
DI_{rjt}^{depth}	-0.000213***	-0.000137**	-0.000215**						
	(0.000077)	(0.000069)	(0.000101)						
$DI_{rjt}^{depth,weighted}$				-0.000167**					
				(0.000073)					
$DI_{rjt}^{depth,LE}$					-0.000128***				
					(0.000036)				
$DI_{rjt}^{depth,WTO+}$						-0.000274***			
						(0.000099)			
$DI_{rjt}^{depth,CoreX}$							-0.000826***		
							(0.000320)		
$DI_{rjt}^{depth,Total}$								-0.000230***	
								(0.000058)	
$DI_{rjt}^{depth,DESTA}$									-0.000454**
									(0.000220)
$DI_{rjt}^{breadth}$	0.000017	-0.000027	0.000017	0.000015	0.000046	0.000016	0.000015	0.000059	-0.000019
	(0.000036)	(0.000037)	(0.000036)	(0.000036)	(0.000039)	(0.000036)	(0.000035)	(0.000040)	(0.000017)
Tariff	-0.012173*	-0.006196	-0.012800*	-0.011788*	-0.014029**	-0.012191*	-0.011708*	-0.013326*	-0.010986
	(0.006865)	(0.006497)	(0.007652)	(0.006870)	(0.006906)	(0.006865)	(0.006858)	(0.006851)	(0.006832)
Baseline *FE and Controls*	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	2,541,619	1,929,538	2,348,033	2,541,619	2,541,619	2,541,619	2,541,619	2,541,619	2,541,619
Adj. R^2	0.326	0.279	0.333	0.326	0.326	0.326	0.326	0.326	0.326

Note: Standard errors clustered at the city–industry level are in parentheses. We include the baseline fixed effects and controls for all the columns. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

exclude this possibility, in addition to controlling for the tariffs, we also focus on the subsample from year 2002 onward, and re-estimate the effect of DI_{rjt}^{depth}. The estimation result is comparable to the baseline estimate, as shown in column (3) of Table 2.

Alternative measures of *DI* — As mentioned before, we compute each country's PTA engagement using a simple average, treating the importance of member countries as equal. As an alternative, we first weight the bilateral PTA depth of destination c with partner n by the share of destination c's exports to partner n in 2000, using IMF trade flows. And then we re-construct DI_{rjt}^{depth} using the weighted PTA engagement, and show the regression results in column (4). The magnitude of the coefficient is comparable with the baseline estimate. We also construct the depth exposure based on the legal enforceability of provisions and the results are presented in column (5). In addition, we use the 14 WTO+ provisions, the 4 core WTO-X provisions, and all the 52 provisions to construct the depth exposure, and the results are shown in columns (6), (7) and (8), respectively. Finally, we construct the depth exposure by using the Design of Trade Agreements Database (DESTA) in column (9).[5] These estimates are stable and robust across different specifications.

[5]The Design of Trade Agreements Database (DESTA) looks in more detail at the content and depth of specific provisions in PTAs (Dür et al., 2014). Specifically, the coding covers a number of aspects for each area ranging from the simple inclusion of a chapter in the PTA to details on national treatment or harmonization of policies. Nevertheless, the coverage is limited to 10 policy areas.

4 Mechanisms

In this section, we explore potential mechanisms through which the deep integration of export destinations with other countries affects the innovation behavior of Chinese firms. One natural mechanism is the Schumpeterian effect (Schumpeter, 1942): the deep integration of export destinations intensifies the competition for Chinese firms and reduces the potential rents that a firm could capture from innovating. This in turn has a negative impact on firm innovation. To examine how competitive conditions change for Chinese firms, we will first focus on how DI impacts the intensive margin of trade. Specifically, we examine the classical trade diversion channel and its decomposition into the price effect and the quantity effect. Then we study the effect of DI at the extensive margin of the trade.

Equally important, the escape-competition effect may serve as a countervailing force. When firms are locked in neck-and-neck competition in the international market, more productive firms will innovate more to escape competition from their rivals (Aghion et al., 2005). Hence, we also study how firms with different initial productivity levels react to DI.

4.1 The Schumpeterian effect of competition

We use detailed Customs data at the firm-product-destination level to study the effect of DI on exports and imports, with products defined at the HS 6-digit level. These data offer the additional advantage of enabling the decomposition of sales into price and quantity, thereby providing insights into the underlying mechanisms.

The empirical specification is as follows:

$$Y_{f,c,p,rjt} = \theta_0 + \theta_1 DI_{rjt}^{depth} + \theta_2 DI_{rjt}^{breadth} + \theta_3 \mathbf{X}_{rjt} + \gamma_{ft} + \lambda_{ct} + \phi_{fcp} + \varepsilon_{f,c,p,rjt}. \tag{4}$$

where $Y_{f,c,p,rjt}$ denotes the logarithm of the trade value, quantity, or price of product p traded between firm f and country c, with the firm located in city r and industry j in year t. γ_{ft} denotes firm-year fixed effects, which absorb firm-level time-varying controls, thanks to the more granular data. The vector $\mathbf{X}_{rjt}$ contains all the baseline controls except for the interactions between firm characteristics and year dummies which are absorbed by the firm-year fixed effects. λ_{ct} denotes country-year fixed effects that control for aggregate shocks specific to each trading partner. We additionally control for

firm-country-product fixed effects ϕ_{fcp}, so that the trade mechanism analysis exploits within-firm-product-destination variation over time.

Columns (1)–(3) of Table 3 report the effects of PTA depth and breadth exposure on exports, while columns (4)–(6) examine imports. Column (1) shows that both the depth and breadth exposures reduce Chinese firms' export values, consistent with trade diversion away from Chinese exporters. Interestingly, however, the two exposures exhibit contrasting patterns when export values are decomposed into quantities and prices, as shown in columns (2) and (3).

The expansion of PTA networks is associated with trade diversion primarily through a contraction in export quantities. Conditional on PTA breadth, deeper integration further reduces Chinese export revenues, but primarily through lower unit prices rather than quantities. Intuitively, deeper integration among export destinations and their PTA partners reduces trade barriers and regulatory costs among members through preferential provisions. As a result, PTA partners can expand their sales in destination markets, driving down prices and diverting trade away from Chinese firms. On the other hand, non-discriminatory provisions benefiting third countries can also reduce trade costs and non-tariff barriers, further lowering prices in destination markets (Mattoo et al., 2022). Chinese exporters may respond by lowering prices sufficiently to expand quantities sold, although the decline in prices more than offsets the increase in quantities, resulting in lower export revenues. Overall, these findings are consistent with stronger competitive pressure operating primarily through price adjustments.

Neither integration measure has a statistically significant effect on total import values. Nevertheless, the decomposition again reveals offsetting adjustments: greater depth lowers import prices, whereas greater breadth reduces import quantities but raises import prices. Notably, the larger decline in export prices relative to import prices suggests a deterioration in China's terms of trade, reminiscent of the theoretical finding that PTAs can reduce third-country welfare and terms of trade (Mossay and Tabuchi, 2015).

Beyond the intensive margin, we also examine the effects of the two DI measures on the extensive margins of exports and imports. Specifically, we use the number of HS6 products each firm exports or imports and the number of export destinations or import source countries as dependent variables. The regression specification is therefore at the firm level. Table 4 reports the results. Greater PTA depth exposure reduces the number of products exported but has no significant effect on the number

Table 3: Trade effect of DI at the firm-product-destination level: Intensive margin

	(1)	(2)	(3)	(4)	(5)	(6)
	Export			Import		
	Value	Quantity	Price	Value	Quantity	Price
DI^{depth}	-0.0019**	0.0033***	-0.0052***	-0.0008	0.0008	-0.0017***
	(0.0009)	(0.0010)	(0.0005)	(0.0007)	(0.0008)	(0.0004)
$DI^{breadth}$	-0.0010**	-0.0027***	0.0017***	-0.0004	-0.0014***	0.0010***
	(0.0004)	(0.0005)	(0.0003)	(0.0004)	(0.0004)	(0.0002)
$\mathbf{X_{rjt}}$	✓	✓	✓	✓	✓	✓
Firm-year FE	✓	✓	✓	✓	✓	✓
Country-year FE	✓	✓	✓	✓	✓	✓
Firm-product-country FE	✓	✓	✓	✓	✓	✓
N	34,328,692	34,328,692	34,328,692	15,678,190	15,678,190	15,678,190
R^2	0.794	0.861	0.946	0.808	0.899	0.937

Note: Standard errors are clustered at the city-industry level. The vector $\mathbf{X}_{rjt}$ contains all the baseline controls except for the interactions between firm characteristics and year dummies. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

of export destinations. In contrast, greater PTA breadth exposure reduces both export product scope and the number of export destinations. On the import side, the PTA depth exposure reduces both imported product scope and the number of source countries, whereas breadth exposure significantly reduces the number of source countries but has no statistically significant effect on the number of imported products.

Tables 3 and 4 substantiate the presence of competition in export destinations, not only along the intensive margins (exporters cut prices in response to depth exposure and cut quantities in response to breadth exposure), but also along the extensive margins (exporters shrink their product lines or exit some export markets), which in turn discourages firms from innovating, consistent with the Schumpeterian effect of competition (Schumpeter, 1942).

4.2 The escape-competition effect

The deep integration of export destinations results in intensified competition. Such competition can reduce the profits of less productive firms and thus hamper their innovation. On the other hand, it can also motivate more productive firms to innovate more in order to escape competition from other firms in export markets.

We explore whether the innovation response to destination integration varies across firms based

Table 4: Trade effect at the firm level: Extensive margin

	(1)	(2)	(3)	(4)
	Export		Import	
	# Products	# Countries	# Products	# Countries
DI^{depth}	-0.1018**	0.0095	-0.1134***	-0.0120***
	(0.0410)	(0.0094)	(0.0297)	(0.0046)
$DI^{breadth}$	-0.0232***	-0.0072***	-0.0069	-0.0050***
	(0.0086)	(0.0023)	(0.0080)	(0.0012)
Firm FE	✓	✓	✓	✓
$\mathbf{X}_{rjt}$	✓	✓	✓	✓
N	1,785,278	1,785,278	1,131,489	1,131,489
R^2	0.789	0.791	0.837	0.747

Note: The vector $\mathbf{X}_{rjt}$ contains all the baseline controls except for the interactions between firm characteristics and year dummies. Standard errors are clustered at the city-industry level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

on their initial productivity. We adopt the method of Levinsohn and Petrin (2003) to estimate firms' total factor productivity (TFP) and group firms into quintiles based on their initial TFP within each industry. Based on the baseline specification in equation (3), we interact the depth exposure with firms' initial productivity quintiles. Specifically, we use the following specification:

$$Y_{f,rjt} = \alpha_0 + \sum_{g=1}^{5} \alpha_1^g Quintile_f^g \times DI_{rjt}^{depth} + \alpha_2 DI_{rjt}^{breadth} + \alpha_3 tariff_{rjt} + FE\ and\ Controls + \varepsilon_{f,rjt}, \tag{5}$$

where $Quintile_f^g$ indicates whether firm f belongs to productivity quintile g based on its initial TFP within its own industry, with integer g ranging from 1 to 5.

Figure 2 demonstrates the heterogeneous innovation effects across TFP quintiles. Firms in the lowest TFP quintile exhibit the greatest negative innovation response to DI_{rjt}^{depth}. Interestingly, the negative effect on innovation is alleviated as productivity increases, with firms in the highest TFP quintile showing the strongest positive innovation response. Figure 3 provides evidence that firms in the lower TFP quintiles experience declines in profits, whereas those in the higher TFP quintiles become more profitable in response to the deep integration of export destinations. Figure A8

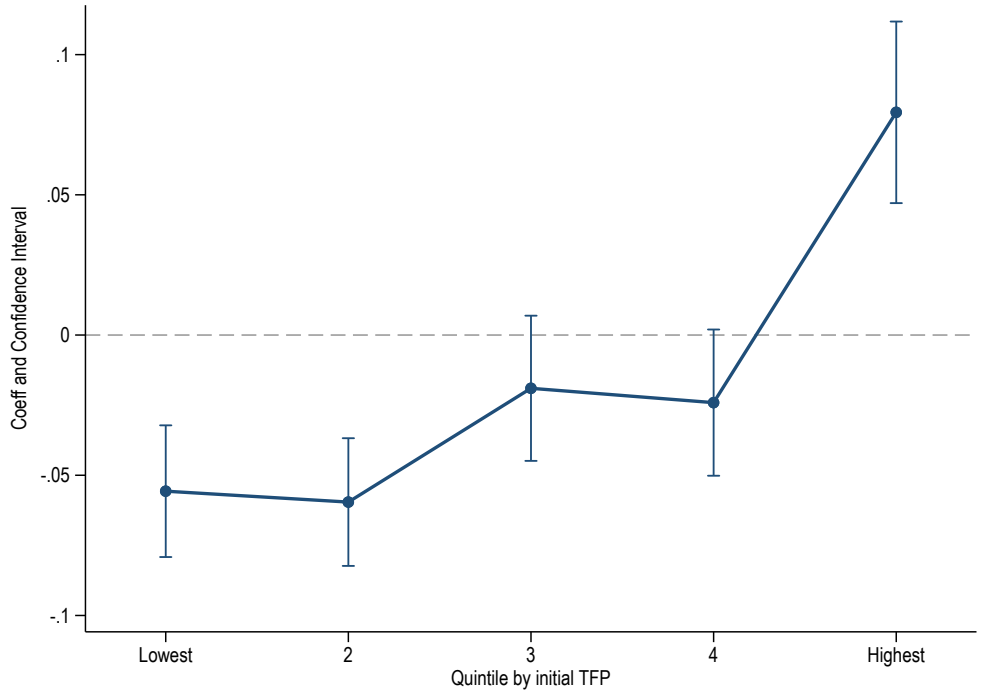


Figure 2: Heterogeneous effects on patenting probability by TFP. The plot shows the coefficient (percentage points) α_1^g of regression (5) for each TFP quintile g and the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is an indicator for whether firm f files any invention patent in year t.

shows that the heterogeneous patterns are robust to using TFP estimated following Ackerberg et al. (2015). Overall, the heterogeneous results for innovation and profits are consistent with both the Schumpeterian effect and the escape-competition effect.

We also find that the escape-competition effect is stronger in R&D intensive sectors.[6] As illustrated in Figure A9, firms in sectors with high R&D intensity have more dispersed responses of innovation to the deep integration of export destinations. This pattern aligns with their profit responses, as shown in Figure A10: compared to firms in low R&D intensity sectors, the most productive firms in high R&D intensity sectors achieve the highest profit gains and the strongest innovation responses. In contrast, the least productive firms in high R&D intensity sectors face the greatest profit losses

[6] Feng et al. (2016) calculate the R&D intensity measures as the ratio of industry R&D costs relative to industry output value for each 4-digit CIC industry. They define the sectoral R&D intensity as the mean value of R&D intensity measures for the 4-digit CIC industries within each 2-digit sector. We take these values and categorize manufacturing sectors into two types: low and high R&D-intensity sectors.

and the weakest innovation performance.

One concern is that deep integration of export destinations may increase exit probabilities among low-productivity firms or induce reallocation toward high-productivity firms. Hence, the heterogeneous pattern may partly reflect compositional changes rather than pure within-firm innovation responses. We examine this concern in two ways. First, Figure A3 repeats the TFP heterogeneity analysis for both patenting probability and profits using a balanced panel of firms observed throughout the sample period. We still observe similar heterogeneous patterns.

Second, we study firm-level entry and exit dynamics in response to destination integration. Specifically, firm entry records a firm's appearance after it was absent in the preceding year. However, firm entry can be volatile, and hence, we also examine first entry, which retains only the first such observed event. Similarly, firm exit records a firm's disappearance in the following year and therefore includes temporary departures; first exit retains only the first observed exit event. Table A6 shows that PTA depth exposure has no significant effect on firm entry or first entry, but significantly increases firm exit and first exit. Because firms enter the industrial survey only after crossing a size threshold, these outcomes describe entry into and exit from the observed sample rather than unambiguously measuring firm birth and death. Entry may partly reflect sales growth that moves firms above the survey cutoff, so exit is more informative about competitive pressure. Overall, although firms may experience higher exit probabilities, our balanced-panel analysis alleviates concerns that the heterogeneous results are driven by compositional changes in the observed firms.

4.3 Additional evidence on competition

We provide additional evidence of the competitive pressure faced by Chinese firms following the deep integration of export destinations. We measure firms' profit rates as the ratio of total profits to gross industrial sales and estimate markups following De Loecker and Warzynski (2012). Using the specification in equation (5), we examine the heterogeneous effects of PTA depth exposure on profit rates and markups across initial productivity quintiles. Interestingly, Figure A6 shows that deeper integration raises profit rates and markups among initially low-productivity firms but lowers both among initially high-productivity firms.

One possible explanation is the compositional adjustment of firms' product portfolios, consistent with the heterogeneous scope-adjustment mechanism in Qiu and Zhou (2013). In their model, glob-

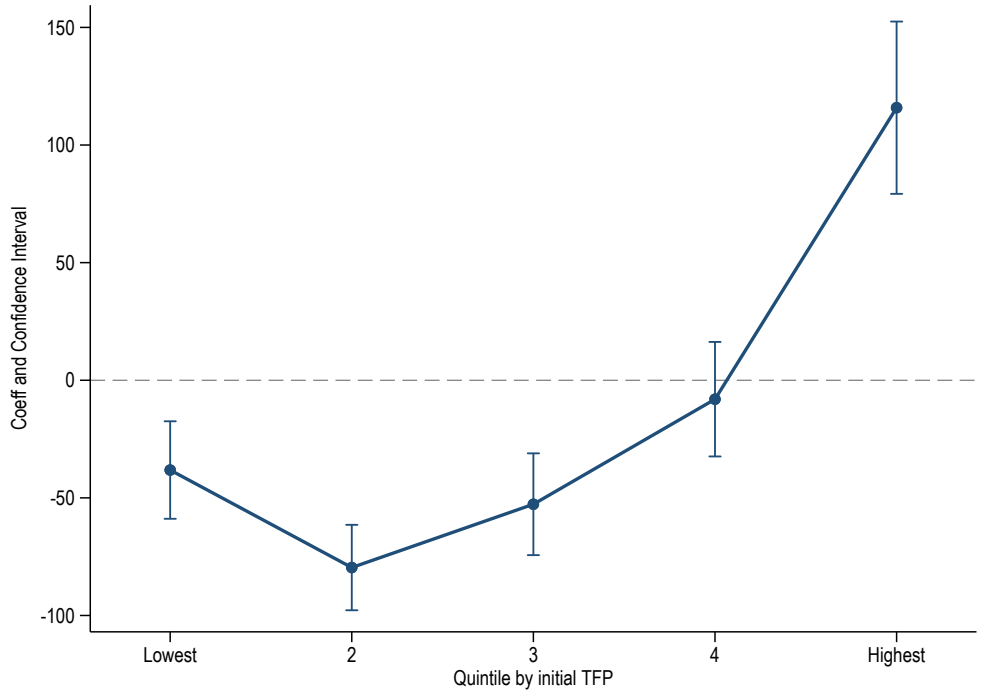


Figure 3: Heterogeneous profit effects by TFP. The plot shows the coefficient α_1^g of regression (5) for each TFP quintile g and the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f's profit level.

alization intensifies competition and reduces markups on surviving products, while firms adjust their product scope differently according to productivity. Less productive firms contract their product range, whereas more productive firms may expand it. As highlighted by Eckel and Neary (2010), products farther from a firm's core competence have higher marginal costs and lower markups. Dropping these peripheral products can therefore raise the average markup of less productive firms despite the direct pro-competitive effect. By contrast, more productive firms may expand into peripheral, lower-markup products, reinforcing the decline in their average markups. This composition channel may also help explain the contrasting responses in firm profit rates.

To corroborate this mechanism, we investigate the heterogeneous effects of destination integration on firms' export scope. Figure A7 shows that less productive firms tend to reduce the number of exported products, whereas more productive firms expand their export product scope. Taken together, the results on profit rates, markups, and export scope are consistent with the heterogeneous scope-adjustment mechanism in Qiu and Zhou (2013), providing additional evidence of competition

faced by Chinese firms arising from the PTA depth exposure.

5 Further heterogeneous effects

5.1 Heterogeneity by PTA provisions

We explore the heterogeneous effects by PTA provisions. Baldwin and Low (2009) classify PTA provisions in two groups: preferential provisions and non-discriminatory (i.e., "most-favoured nation" or MFN) provisions. The former provisions are those that apply only to the countries that signed the PTA, whereas the latter provisions are applied on an MFN basis. Preferential provisions are more likely to accrue benefits exclusively to the member countries, inflicting a further competitive disadvantage on third countries. The resulting trade diversion effect can lead to a negative innovation effect. On the other hand, MFN provisions can generate positive spillovers to third countries (Mattoo et al., 2022; Lee et al., 2023), which can lead to a positive innovation effect.

We classify the core provisions in PTAs into preferential provisions denoted as {Pref}, and MFN provisions, denoted as {MFN}, based on Table A9. Then we focus on legally enforceable provisions and decompose the core depth of each PTA into two types: $\text{CoreDepth}_t^{Pref} = \sum_{i \in \text{Pref}} \text{provision}_{it}^v$, and $\text{CoreDepth}_t^{MFN} = \sum_{i \in \text{MFN}} \text{provision}_{it}^v$. Following Subsection 2.2, we can similarly define DI_{rjt}^{Pref} based on preferential provisions and DI_{rjt}^{MFN} based on MFN provisions.

To study the heterogeneous effect of provisions in the presence of escape-competition effect, we interact DI_{rjt}^{Pref} and DI_{rjt}^{MFN} with firms' initial productivity quintiles, respectively. Specifically, we use the following specification:

$$\begin{aligned} Y_{f,rjt} = \quad & \alpha_0 + \sum_{g=1}^{5} \alpha_1^g Quintile_f^g \times DI_{rjt}^{Pref} \quad + \sum_{g=1}^{5} \alpha_2^g Quintile_f^g \times DI_{rjt}^{MFN} \\ & + \alpha_3 DI_{rjt}^{breadth} + \alpha_4 tariff_{rjt} + FE \text{ and } Controls + \varepsilon_{f,rjt}. \end{aligned} \tag{6}$$

The innovation responses along the initial productivity dimension to DI_{rjt}^{Pref} and DI_{rjt}^{MFN} are plotted in Figure 4. The left panel shows that negative innovation effects are concentrated among the low-productivity firms (the second TFP quintile, Q2), while more productive firms exhibit negative but statistically insignificant responses. This pattern reflects the interaction of two opposing forces.

On the one hand, more productive firms have stronger incentives to innovate in order to escape competition. On the other hand, preferential provisions generate trade diversion for the third country China, which dampens innovation incentives of Chinese firms. For highly productive firms, these forces largely offset each other, whereas for low-productivity firms the weaker escape-competition effect amplifies the trade-diversion channel, resulting in negative innovation responses.

The right panel in Figure 4 shows a significantly positive innovation response among the most productive firms. The large magnitude of this effect is consistent with positive spillovers from MFN provisions to third countries, such as China, reinforced by strong escape-competition incentives at the top of the productivity distribution. In contrast, the least productive firms experience negative patenting propensity, indicating that the potential spillovers from MFN provisions are dominated by their limited ability to innovate in response to competition.[7]

[7]The negative innovation effects for the least productive firms are consistent with Chen and Mattoo (2008), who argue that although MFN provisions may generate scale economies, non-member countries can experience export losses when restrictive rules of origin and the raised stringency of standards exclude them from these benefits.

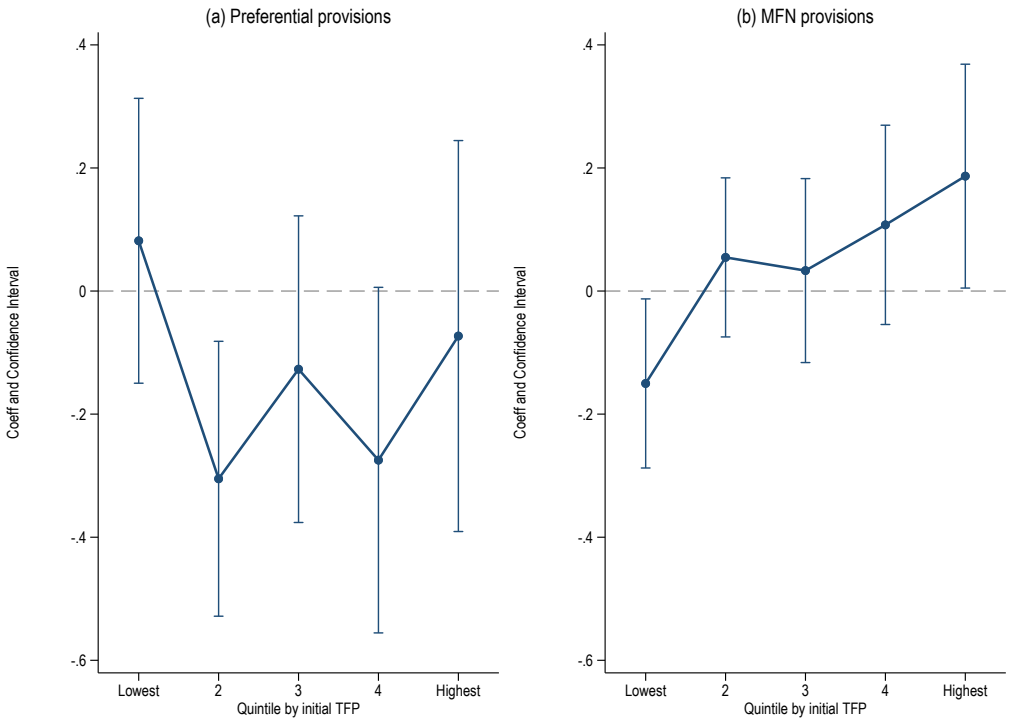


Figure 4: Heterogeneous innovation effects by PTA provisions. DI_{rjt}^{Pref} and DI_{rjt}^{MFN} are constructed based on legally enforceable preferential and MFN provisions, respectively. We interact DI_{rjt}^{Pref} and DI_{rjt}^{MFN} with firms' initial productivity quintiles, respectively, as in equation (6). The plots show the coefficients (percentage points) α_1^g and α_2^g for each TFP quintile g, respectively, with the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is an indicator for whether firm f files any invention patent in year t.

5.2 Heterogeneity by China's PTA partner vs non-partner countries

In this subsection, we analyze how China's PTA membership mediates the innovation effect of the deep integration of export destinations. Intuitively, destinations that are China's PTA partners (as represented by country B in Figure 5), on average, tend to be more competitive for Chinese firms, compared to those that are not China's PTA partners (as represented by country A in Figure 5), conditional on the same deep integration with other countries. The competition induced by PTA links between China and its export destinations may amplify innovation responses, particularly for firms at both ends of the productivity spectrum. Specifically, we expect that more productive firms will innovate more and less productive firms will innovate less, in response to DI when exporting to

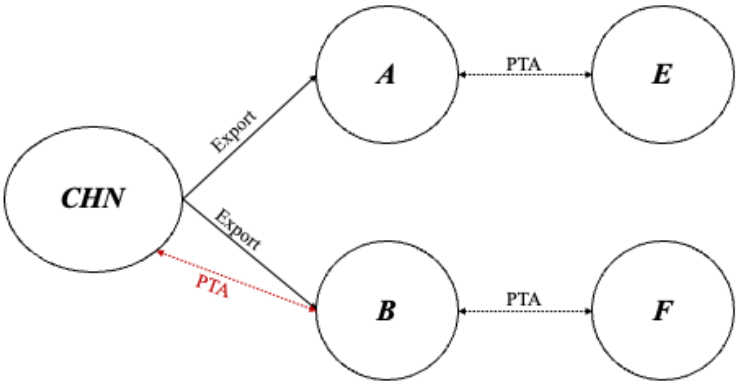


Figure 5: China's PTA partner country B and non-partner country A.

China's PTA partners.

We first classify export destinations into China's PTA partners, denoted as {P}, and non-PTA partner countries, denoted as {NP}, based on whether China has ever signed PTAs with these countries. Then we compute the deep integration of the two types of export destinations, respectively. Specifically, the deep integration of China's PTA partners $DI_{rjt}^{depth,P}$ is defined as follows:

$$DI_{rjt}^{depth,P} = \sum_{c \in \{\text{P}\}} Share_{rjc,0}\, DepthEngage_{ct},$$

And the deep integration of China's non-PTA partners $DI_{rjt}^{depth,NP}$ is defined as follows:

$$DI_{rjt}^{depth,NP} = \sum_{c \in \{\text{NP}\}} Share_{rjc,0}\, DepthEngage_{ct}.$$

We interact $DI_{rjt}^{depth,P}$ and $DI_{rjt}^{depth,NP}$ with firms' initial productivity quintiles, respectively. Specifically, we use the following specification:

$$\begin{aligned} Y_{f,rjt} = \alpha_0 + \sum_{g=1}^{5} \alpha_1^g Quintile_f^g \times DI_{rjt}^{depth,P} + \sum_{g=1}^{5} \alpha_2^g Quintile_f^g \times DI_{rjt}^{depth,NP} \\ + \alpha_3 DI_{rjt}^{breadth} + \alpha_4 tariff_{rjt} + FE\ and\ Controls + \varepsilon_{f,rjt}. \end{aligned} \tag{7}$$

The innovation responses along the initial productivity dimension to $DI_{rjt}^{depth,P}$ and $DI_{rjt}^{depth,NP}$

are plotted in Figure 6. As shown in Panel (a), we find a stronger escape-competition effect for firms that export to China's PTA partners, i.e., a wider dispersion of innovation responses to the deep integration of China's PTA partners, compared with that of China's non-PTA partners in Panel (b).

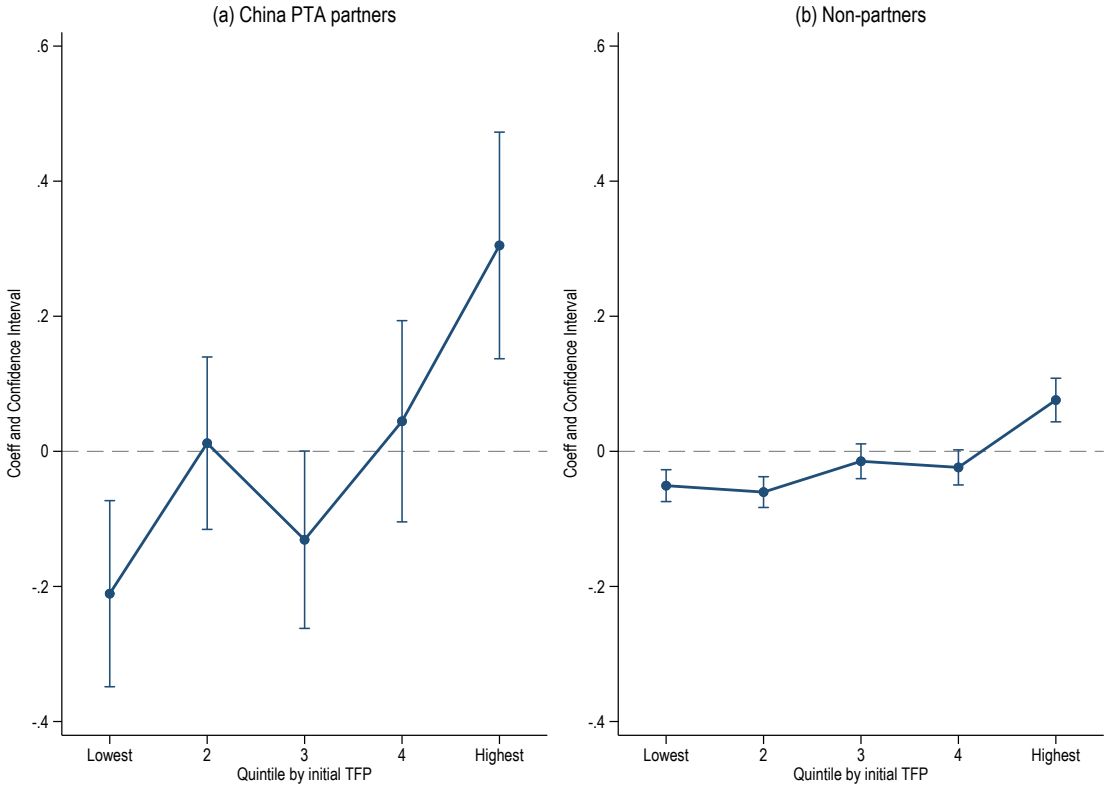


Figure 6: Heterogeneous innovation effects by China's PTA partnership. $\text{DI}_{rjt}^{depth,P}$ and $\text{DI}_{rjt}^{depth,NP}$ are constructed based on China's PTA partners and non-partners, respectively. We interact $\text{DI}_{rjt}^{depth,P}$ and $\text{DI}_{rjt}^{depth,NP}$ with firms' initial productivity quintiles, respectively, as in equation (7). The plots show the coefficients (percentage points) α_1^g and α_2^g for each TFP quintile g, respectively, with the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is an indicator for whether firm f files any invention patent in year t.

5.3 Innovation type heterogeneity

We study whether the impact of the deep integration of export destinations on Chinese firms' innovation varies by patent types: invention, utility, and design patents.

Table 5 reports the heterogeneous impacts. Column (1) shows the effect on invention patents

Table 5: Heterogeneity results by patent types

	(1)	(2)	(3)
	Patent types		
	Invention	Utility	Design
DI^{depth}	-0.000213***	-0.000147	0.000031
	(0.000077)	(0.000090)	(0.000059)
$DI^{breadth}$	0.000017	0.000012	0.000022
	(0.000036)	(0.000040)	(0.000026)
N	2,541,619	2,541,619	2,541,619

Note: Each dependent variable equals one if the firm files a positive number of patents of the indicated type in year t. All regressions include baseline controls. Standard errors are clustered at the city–industry level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

for comparison. Column (2) shows that PTA depth exposure has a negative, albeit statistically insignificant, effect on the probability of filing utility patents. This finding may be consistent with the learning-from-exporting mechanism documented by Liang et al. (2024). As also highlighted by Liu et al. (2021), utility patenting may benefit more from knowledge spillovers through imitation. The decline in exports induced by deeper integration of export destinations may therefore limit firms' opportunities to learn from foreign markets, thereby discouraging utility innovation. Column (3) indicates no correlation between design patent filings and the deep integration of export destinations. This may suggest a limited role of design patents in competing for international markets, especially if the competition is more related to technological advances or cost-efficiency rather than aesthetics (appearance or shape).

5.4 Other dimensions

We further explore heterogeneity along three additional dimensions: industry upstreamness, local IPR enforcement, and firm ownership. First, the negative innovation responses among less productive firms tend to be stronger in upstream industries, while the positive responses among more productive firms tend to be weaker. This pattern is consistent with upstream industries being more exposed to trade diversion, as PTA provisions such as rules of origin may shift intermediate-input sourcing from third countries toward PTA members. Second, stronger local IPR enforcement amplifies productivity heterogeneity, with more productive firms increasing innovation and less productive firms reducing

innovation in response to PTA depth exposure. Finally, the negative effect on patenting is statistically significant only among private firms, although the estimates are negative across all ownership types. We relegate these additional heterogeneity analyses to the Appendix Section A4.

6 Technological direction of invention patenting

We have shown that deeper integration among export destinations reduces firms' patenting propensity, while its effect on patenting counts is negative but insignificant. Beyond these effects, destination integration may also shape the technological direction of innovation. As Lim et al. (2022) highlight, trade-induced changes in market size and competition across quality segments can shape firms' innovation incentives. In our setting, PTAs among export destinations may reshape the relative competitive conditions across markets through preferential market access, regulatory cooperation, and product standards. Firms serving multiple export destinations, potentially facing different preferences and standards, may therefore have stronger incentives to reallocate innovation across technological fields rather than simply adjusting its scale. To explore this possibility, we examine whether destination integration affects the technological composition of firms' invention patenting and whether this response differs between firms serving single and multiple export destinations.

We use IPC4 codes to examine whether firms redirect the technological composition of their invention patenting. Let n_{ftk} denote the number of invention patents filed by firm f in year t in IPC4 class k, and let t_f^0 denote the firm's first year of invention patenting. We define the firm's initial technological portfolio as $\mathbf{p}_{f0} = (p_{f0k})_k$, where $p_{f0k} = n_{ft_f^0 k} / \sum_\ell n_{ft_f^0 \ell}$. In any subsequent patenting year, the current portfolio is $\mathbf{p}_{ft} = (p_{ftk})_k$, where $p_{ftk} = n_{ftk} / \sum_\ell n_{ft\ell}$. All vectors are defined over the union of IPC4 classes observed in the firm's initial and current portfolios, with a zero share assigned to classes absent from a given portfolio.

We construct four measures of innovation direction. First, *Any new IPC* is defined as $\mathbb{I}\{\sum_{k:p_{f0k}=0} n_{ftk} > 0\}$, indicating whether the firm files at least one invention patent in an IPC4 class absent from its initial technological portfolio. Second, the *New-IPC share* is defined as $\sum_{k:p_{f0k}=0} n_{ftk} / \sum_k n_{ftk}$, measuring the share of current invention patents belonging to new IPC4 classes. Third, *IPC distance* is defined as one minus the cosine similarity between the current and initial IPC-share vectors, $IPC\ Distance_{ft} = 1 - \frac{\sum_k p_{f0k} p_{ftk}}{\sqrt{\sum_k p_{f0k}^2}\sqrt{\sum_k p_{ftk}^2}}$. This measure ranges from zero to one, with larger values

indicating greater technological reorientation. Unlike the New-IPC share, IPC distance also captures reallocations among IPC4 classes already present in the firm's initial portfolio. Fourth, *Top IPC changed* is defined as $\mathbb{I}\{\arg\max_k p_{ftk} \neq \arg\max_k p_{f0k}\}$, indicating whether the firm's leading IPC4 class differs from that in its initial portfolio. Firm-years before the initial invention-patenting year and firms that never establish an invention-patent portfolio cannot be assigned a direction measure and are excluded.

Table 6 reports the average effects. Greater PTA depth exposure significantly reduces the probability of introducing any new IPC class, the share of patents in new IPC classes, and the distance from the initial technological portfolio. The effect on whether the leading IPC class changes is also negative but statistically insignificant. These results suggest that deeper destination integration is associated with less technological reorientation on average, in addition to the decline in overall invention patenting documented earlier.

The response of innovation direction to destination integration may depend on the number of export markets a firm serves. Firms exporting to multiple destinations are exposed to potentially heterogeneous consumer preferences, product standards, and competitive conditions. As PTAs reshape competition across these markets, such firms may have greater incentives and opportunities to redirect innovation toward technologies better suited to different destination markets. By contrast, firms serving a single export destination have more limited opportunities for reallocating innovation across markets. We therefore investigate whether the innovation-direction response to PTA depth exposure differs between single- and multiple-destination exporters.

Table 7 reports the results. All four interaction coefficients are positive, with statistically significant effects on the New-IPC share and IPC distance at the 5% level and on the change in the leading IPC class at the 10% level. The interaction for Any new IPC is also positive although statistically insignificant. These results suggest that, relative to single-destination exporters, firms serving multiple destinations exhibit a more positive response in the technological direction of innovation to deeper destination integration. This pattern is consistent with multi-market firms confronting more heterogeneous preferences or product standards and therefore having greater scope to adjust the composition of innovation.

Table 6: Destination integration and the direction of invention patenting

	Any new IPC	New-IPC share	IPC distance	Top IPC changed
DI_{rjt}^{depth}	-0.002681**	-0.002582**	-0.002432**	-0.001953
	(0.001363)	(0.001206)	(0.001197)	(0.001324)
$DI_{rjt}^{breadth}$	0.000507	0.000423	0.000429	0.000581*
	(0.000325)	(0.000285)	(0.000284)	(0.000315)
Observations	107,926	107,926	107,926	107,926

Notes: The sample covers all firms that have invention patents. Firm-years preceding a firm's initial invention-patenting year are excluded, whereas all subsequent firm-years are retained, including those with no invention-patent filings. All vectors are defined over the union of IPC4 classes observed in the firm's initial and current portfolios, with a zero share assigned to classes absent from a given portfolio. All regressions include the baseline controls. Standard errors clustered at the city–industry level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Export-market scope and the direction of invention patenting

	Any new IPC	New-IPC share	IPC distance	Top IPC changed
DI^{depth}	-0.003678	-0.009759**	-0.010080**	-0.008241
	(0.005498)	(0.004692)	(0.004571)	(0.005130)
$DI^{depth} \times$ *Multiple_destinations*	0.005903	0.008765**	0.007937**	0.007468*
	(0.004601)	(0.003971)	(0.003792)	(0.004288)
$DI^{breadth}$	0.000442	0.000348	0.000448	0.000570
	(0.000726)	(0.000630)	(0.000622)	(0.000698)
Observations	25,996	25,996	25,996	25,996

Notes: The sample covers all exporters that have invention patents. Firm-years before the initial invention-patenting year and firms that never establish an invention-patent portfolio cannot be assigned a direction measure and are excluded. All vectors are defined over the union of IPC4 classes observed in the firm's initial and current portfolios, with a zero share assigned to classes absent from a given portfolio. *Multiple_destinations* equals one when a firm exports to more than one foreign country and zero when it exports to exactly one country. All regressions include the baseline controls. Standard errors clustered at the city–industry level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

7 Conclusion

This paper examines how deep PTA integration among export destinations affects innovation in non-member countries. We approach this research question from the perspective of China as a third country. We construct two shift-share measures of destination integration based on the depth and breadth of PTAs engaged in by export destinations. We find that deeper integration reduces invention patenting primarily along the extensive margin, while the PTA breadth exposure has no significant effect. The PTA depth exposure also reduces technological reorientation on average, but firms serving multiple export destinations exhibit a more positive response than single-destination exporters, consistent with their greater scope to adjust innovation across markets with heterogeneous preferences or product standards.

The Schumpeterian effect and the escape-competition effect are potential mechanisms at work through which destination integration affects the innovation behavior of Chinese firms. The PTA depth exposure reduces export values and prices, reflecting the trade diversion effect, a driving force of the negative innovation effect. Meanwhile, exporters export fewer products. On the other hand, we find the least productive firms suffer profit losses and exhibit the greatest negative innovation response to destination integration. Interestingly, the negative effects are alleviated for more productive firms. The most productive firms achieve the highest profits and the strongest innovation responses, in line with the escape-competition effect. The escape-competition effect is stronger in sectors with higher R&D intensity. We also show that firms' innovation responses to the deep integration of export destinations vary by PTA provisions, China's PTA partnership, firms' patent types, upstreamness, local IPR enforcement and firm ownership.

The results in this paper have important policy implications. The heterogeneous innovation responses across the TFP spectrum highlight the importance of strengthening domestic IPR enforcement to safeguard and promote firm-level innovation. Enhanced IPR enforcement would foster a more supportive environment for firms to innovate and develop competitive technologies, enabling them to better navigate the competition in export markets undergoing deep integration through PTAs. This is particularly crucial for highly productive firms operating in R&D-intensive sectors or exporting to destinations that are China's PTA partners. On the other hand, proactively expanding PTA networks, strengthening MFN provisions (such as IPR), and facilitating intermediate-goods trade are complementary and feasible policy strategies. This dual approach aligns economic growth

with innovation-driven development, offering an effective policy pathway for sustainable economic advancement.

Our findings also have welfare implications. Innovation is an important source of economic growth. However, in response to the deep integration of export destinations, Chinese firms experienced a decline in invention patenting on average. Meanwhile, the relatively larger decline in export prices against import prices suggests a deterioration of the terms of trade. Both seem to suggest a potential welfare loss for China. Future research could focus on developing a structural model to assess the overall third-country welfare impact, accounting for the heterogeneous effects on innovation and technological reorientation.

References

Ackerberg, D. A., Caves, K., and Frazer, G. (2015). Identification properties of recent production function estimators. *Econometrica*, 83(6):2411–2451.

Aghion, P., Bergeaud, A., Gigout, T., Lequien, M., and Melitz, M. (2023). Exporting ideas: Knowledge flows from expanding trade in goods.

Aghion, P., Bergeaud, A., Lequien, M., and Melitz, M. J. (2024). The heterogeneous impact of market size on innovation: Evidence from French firm-level exports. *Review of Economics and Statistics*, pages 1–56.

Aghion, P., Bloom, N., Blundell, R., Griffith, R., and Howitt, P. (2005). Competition and innovation: An inverted-U relationship. *Quarterly Journal of Economics*, 120(2):701–728.

Antràs, P., Chor, D., Fally, T., and Hillberry, R. (2012). Measuring the upstreamness of production and trade flows. *American Economic Review*, 102(3):412–416.

Autor, D. H., Dorn, D., and Hanson, G. H. (2013). The China syndrome: Local labor market effects of import competition in the United States. *American Economic Review*, 103(6):2121–2168.

Aw, B. Y., Roberts, M. J., and Xu, D. Y. (2011). R&D investment, exporting, and productivity dynamics. *American Economic Review*, 101(4):1312–1344.

Baier, S. L. and Bergstrand, J. H. (2007). Do free trade agreements actually increase members' international trade? *Journal of International Economics*, 71(1):72–95.

Baier, S. L., Yotov, Y. V., and Zylkin, T. (2019). On the widely differing effects of free trade agreements: Lessons from twenty years of trade integration. *Journal of International Economics*, 116:206–226.

Baldwin, R. and Low, P. (2009). *Multilateralizing regionalism: challenges for the global trading system*. Cambridge University Press.

Baltagi, B. H., Egger, P., and Pfaffermayr, M. (2008). Estimating regional trade agreement effects on fdi in an interdependent world. *Journal of Econometrics*, 145(1-2):194–208.

Bartik, T. J. (1991). *Who Benefits from State and Local Economic Development Policies?* W.E. Upjohn Institute for Employment Research, Kalamazoo, MI.

Borusyak, K. and Hull, P. (2023). Nonrandom exposure to exogenous shocks. *Econometrica*, 91(6):2155–2185.

Borusyak, K., Hull, P., and Jaravel, X. (2022). Quasi-experimental shift-share research designs. *Review of Economic Studies*, 89(1):181–213.

Borusyak, K., Hull, P., and Jaravel, X. (2025). A practical guide to shift-share instruments. *Journal of Economic Perspectives*, 39(1):181–204.

Bustos, P. (2011). Trade liberalization, exports, and technology upgrading: Evidence on the impact of mercosur on argentinian firms. *American Economic Review*, 101(1):304–340.

Campi, M. and Dueñas, M. (2019). Intellectual property rights, trade agreements, and international trade. *Research Policy*, 48(3):531–545.

Carr, J. B. and Packham, A. (2019). Snap benefits and crime: Evidence from changing disbursement schedules. *Review of Economics and Statistics*, 101(2):310–325.

Chen, M. X. and Mattoo, A. (2008). Regionalism in standards: good or bad for trade? *Canadian Journal of Economics/Revue canadienne d'économique*, 41(3):838–863.

Conconi, P., García-Santana, M., Puccio, L., and Venturini, R. (2018). From final goods to inputs: the protectionist effect of rules of origin. *American Economic Review*, 108(8):2335–2365.

Dai, M., Yotov, Y. V., and Zylkin, T. (2014). On the trade-diversion effects of free trade agreements. *Economics Letters*, 122(2):321–325.

De Loecker, J. and Warzynski, F. (2012). Markups and firm-level export status. *American Economic Review*, 102(6):2437–2471.

Dür, A., Baccini, L., and Elsig, M. (2014). The design of international trade agreements: Introducing a new dataset. *Review of International Organizations*, 9:353–375.

Dutt, P. (2020). The WTO is not passé. *European Economic Review*, 128:103507.

Eckel, C. and Neary, J. P. (2010). Multi-product firms and flexible manufacturing in the global economy. *Review of Economic Studies*, 77(1):188–217.

Felbermayr, G. J. and Teti, F. A. (2023). Revisiting the trade-creating effects of non-tariff barriers. Policy Research Working Paper 10322, World Bank.

Feng, L., Li, Z., and Swenson, D. L. (2016). The connection between imported intermediate inputs and exports: Evidence from chinese firms. *Journal of International Economics*, 101:86–101.

He, Q. and Luo, X. (2026). Catalysts for third-country market entry: The role of preferential trade agreements. *World Development*, 205:107437.

Hofmann, C., Osnago, A., and Ruta, M. (2017). Horizontal depth: a new database on the content of preferential trade agreements. *World Bank Policy Research Working Paper*, (7981).

Kogan, L., Papanikolaou, D., Seru, A., and Stoffman, N. (2017). Technological innovation, resource allocation, and growth. *Quarterly Journal of Economics*, 132(2):665–712.

Lee, W., Mulabdic, A., and Ruta, M. (2023). Third-country effects of regional trade agreements: A firm-level analysis. *Journal of International Economics*, 140:103688.

Levinsohn, J. and Petrin, A. (2003). Estimating production functions using inputs to control for unobservables. *Review of Economic Studies*, 70(2):317–341.

Liang, Y., Shi, K., Tao, H., and Xu, J. (2024). Learning by exporting: Evidence from patent citations in china. *Journal of International Economics*, 150:103933.

Lileeva, A. and Trefler, D. (2010). Improved access to foreign markets raises plant-level productivity... for some plants. *Quarterly Journal of Economics*, 125(3):1051–1099.

Lim, K., Trefler, D., and Yu, M. (2022). Looking backward, innovating forward: A theory of competitive cascades. NBER Working Paper 30455, National Bureau of Economic Research.

Liu, Q., Lu, R., Lu, Y., and Luong, T. A. (2021). Import competition and firm innovation: Evidence from China. *Journal of Development Economics*, 151:102650.

Liu, Q. and Ma, H. (2020). Trade policy uncertainty and innovation: Firm level evidence from China's wto accession. *Journal of International Economics*, 127:103387.

Luo, X. and Hsu, W.-T. (2024). Market size and institution in a global economy. *Available at SSRN 4833286*.

Maskus, K. E. and Ridley, W. (2021). *Trade Impacts of Intellectual-Property-Related PTAs: Evidence from Using the World Bank Deep Trade Agreements Database*. The World Bank.

Mattoo, A., Mulabdic, A., and Ruta, M. (2022). Trade creation and trade diversion in deep agreements. *Canadian Journal of Economics/Revue canadienne d'économique*, 55(3):1598–1637.

Medvedev, D. (2012). Beyond trade: the impact of preferential trade agreements on FDI inflows. *World Development*, 40(1):49–61.

Melitz, M. J. and Redding, S. J. (2021). Trade and innovation. Technical report, National Bureau of Economic Research.

Mossay, P. and Tabuchi, T. (2015). Preferential trade agreements harm third countries. *Economic Journal*, 125(589):1964–1985.

Panagariya, A. (2000). Preferential trade liberalization: the traditional theory and new developments. *Journal of Economic literature*, 38(2):287–331.

Qiu, L. D. and Zhou, W. (2013). Multiproduct firms and scope adjustment in globalization. *Journal of International Economics*, 91(1):142–153.

Schumpeter, J. A. (1942). *Capitalism, socialism and democracy*. Routledge.

Shu, P. and Steinwender, C. (2019). The impact of trade liberalization on firm productivity and innovation. *Innovation Policy and Economy*, 19(1):39–68.

Teti, F. A. (2024). Missing tariffs. CESifo Working Paper 11590, CESifo.

Viner, J. (1950). *The customs union issue*. New York: Carnegie Endowment for International Peace.

Wang, X., Fan, G., and Yu, J. (2016). Chinese provinces marketization index report. *Social Sciences Academic Press*.

Yu, M. (2015). Processing trade, tariff reductions and firm productivity: Evidence from chinese firms. *Economic Journal*, 125(585):943–988.

Appendix

A1 Data matching strategy and summary statistics

We first concord the CIC 4-digit industries with HS 6-digit product codes. To this end, we first convert all HS 6-digit product codes to those of the HS96 version, using the conversion tables provided by WITS that map between HS96, HS02, HS07, and HS12. Then, we map CIC 4-digit industries from the 2011 version to 2002 version. And finally, we map HS96 6-digit products to the CIC 4-digit industries, based on the concordance table from the National Bureau of Statistics of China.

Our matching strategy involves two steps. Firstly, we merge the patent filing dataset and ASIF by the firm name reported in both datasets (Liu et al., 2021). The matched data have an unbalanced panel of around 958,636 firms and a total of around 4,085,073 observations, with detailed patent filing information and firm characteristics between 2000-2013.[8]

Secondly, we merge the *DI* dataset obtained in subsection 2.2 with the matched dataset in the first step. The sample includes all firms within city-industry pairs that recorded exports in the year 2000. Note that around 56% of the city-industry pairs do not export in the sample in 2000, and hence firms in these pairs will not be taken into account in the analysis.[9] Eventually, the final matched dataset consists of 648,325 firms and 2,745,950 observations. The sample used in our study covers the period from 2000 to 2013. Table A1 reports the summary statistics.

[8]The matching quality resembles that of Liu et al. (2021). We have a similar number of firms, observations and patent filings if we focus on the same period 1998-2005 as in Liu et al. (2021).

[9]Yu (2015) directly matched the ASIF dataset with the Customs dataset, and around 13.6% of manufacturing firms remain in the dataset. In contrast to his method, our matching in the second step keeps firms that do not export but are still affected by the DI shocks at the city-industry level.

Table A1: Summary Statistics

Variables	N	Mean	S.D.	Min	Max
DI^{depth}	2,745,950	11.91	3.96	0.00	18.00
$DI^{breadth}$	2,745,950	20.42	13.42	0.00	82.00
Invention patenting, extensive margin	2,745,950	0.03	0.18	0.00	1.00
Invention patenting, intensive margin	91,035	0.82	0.97	0.00	8.67
Tariff	2,745,950	0.04	0.06	0.00	1.55
Log firm age	2,745,509	3.15	0.31	2.56	4.20
Log employment	2,730,054	4.82	1.13	1.79	7.79
Log capital–labor ratio	2,729,967	5.14	1.14	-1.50	12.70
Foreign firm	2,745,950	0.13	0.34	0.00	1.00
Private firm	2,745,950	0.79	0.41	0.00	1.00
Exporting firm	2,231,231	0.33	0.47	0.00	1.00

Notes: The sample includes all firms within city-industry pairs that recorded exports in the year 2000. DI^{depth} is exposure to the average depth of destination-country PTA engagement, and $DI^{breadth}$ is exposure to the number of destination-country PTA partners, excluding China. The extensive margin equals one when a firm files at least one invention patent in a year. The intensive margin is the logarithm of invention patent filings and is defined only for positive filings. Data Sources: Authors' own calculation based on merged datasets from China National Bureau of Statistics, China's State Intellectual Property Office (SIPO), Annual Survey of Industrial Firms (ASIF), China's Customs, and the World Bank Deep Trade Agreements Database.

A2 Identifying assumption checks

In this section, we address potential endogeneity concerns in our shift-share design. We complement the baseline results with additional diagnostic tests and robustness checks to assess the plausibility of the identifying assumption.

Exogeneity of shifts and shares — Following the practical guide by Borusyak et al. (2025), we examine potential threats to exogeneity at both the shift and share levels. At the shift level, we have already controlled for destination market size to isolate the effects of inter-connectedness among export destinations from those of destination demand. Nevertheless, the depth variable may also capture

other factors, such as institutional spillovers or regional demand conditions. The correlation between destination-level PTA depth shifts and five-year-lagged leave-one-out regional Rule of Law is 0.166, while the corresponding correlation with leave-one-out regional GDP growth is -0.081.[10] To account for these potential confounding factors, we construct Bartik-type controls for both regional variables by weighting their shift-level values using the same initial export shares as in our DI measures.

For the share-side tests, we focus on the five export destinations with the largest absolute Rotemberg weights: Japan, Korea, the United States, Indonesia, and Malaysia. For each destination, we conduct a balance test to examine whether the initial city–industry export shares in 2000 are systematically correlated with baseline city–industry characteristics. These characteristics include the initial share of patenting firms, log total sales, the foreign-firm sales share, and the private-firm sales share. Across the five regressions, log total sales and the foreign-firm sales share are statistically significant in all cases, while the other two characteristics are significant only in some specifications. We therefore further control for the interactions between these four initial characteristics and year dummies.

Table A2 reports the results. Column (1) introduces the shift-level controls, Column (2) includes the share-level controls, and Column (3) incorporates both sets of controls. The estimated coefficients on DI depth are -0.000195, -0.000172, and -0.000179, respectively. The first two estimates are statistically significant at the 5% level, while the third is significant at the 10% level. All three estimates are comparable in magnitude to the baseline estimate. The coefficients on DI breadth remain statistically insignificant across all specifications. Overall, these results suggest that our baseline findings are robust to controlling for potential confounding factors at both the shift and share levels.

[10] The Rule of Law index is taken from the Worldwide Governance Indicators compiled by the World Bank.

Table A2: Shift- and share-side exogeneity diagnostics

	(1)	(2)	(3)
	Shift controls	Share controls	Both controls
DI_{rjt}^{depth}	-0.000195**	-0.000172**	-0.000179*
	(0.000093)	(0.000081)	(0.000092)
$DI_{rjt}^{breadth}$	0.000069	0.000055	0.000071
	(0.000044)	(0.000041)	(0.000044)
Regional Rule of Law exposure	✓		✓
Regional GDP-growth exposure	✓		✓
Initial patenting-firm share × year		✓	✓
Initial log total sales × year		✓	✓
Initial foreign sales share × year		✓	✓
Initial private sales share × year		✓	✓
Baseline controls and fixed effects	✓	✓	✓
Observations	2,258,162	2,258,162	2,258,162

Notes: The dependent variable is an indicator for positive invention-patent filings. Regional institutional quality and regional GDP growth exclude the destination country itself and vary by year. Share-side characteristics are measured in 2000 and receive unrestricted year-specific slopes. The same estimation sample is used in all three columns. All columns otherwise use the baseline controls, fixed effects, and clustering. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Additionally, following Borusyak et al. (2022), we implement the exposure-robust inference using the shift-level equivalence approach. The results are reported in Table A3. Column (1) presents the coefficient estimates and standard errors for DI_{rjt}^{depth} and $DI_{rjt}^{breadth}$ from the baseline firm-level specification in equation (3). We then residualize the outcome and both exposure measures with respect to the same fixed effects and controls, and aggregate the residualized variables to the destination-year shift level using initial export shares through the `ssaggregate` procedure. At the shift level, the aggregated residualized depth and breadth exposures are instrumented by their underlying shifts, DepthEngage_{ct} and $\text{BreadthEngage}_{ct}$, respectively. We consider two alternative clustering schemes to account for potential correlation in shifts: clustering at the destination level in Column (2) and at the geographic-region level in Column (3).[11] The results remain robust across both clustering schemes,

[11] We classify countries into 16 geographic regions: North Africa and Sub-Saharan Africa; the Caribbean, Central

providing further support for our findings under exposure-robust inference.

Table A3: Unit-level and shift-level equivalence

	(1)	(2)	(3)
	Unit level	Shift level	
	Baseline	Destination cluster	Region cluster
DI^{depth}	-0.000213***	-0.000213**	-0.000213***
	(0.000077)	(0.000088)	(0.000077)
$DI^{breadth}$	0.000017	0.000017	0.000017
	(0.000036)	(0.000047)	(0.000026)
Observations	2,541,619	2,643	2,643

Notes: Standard errors are reported in parentheses. Column (1) reports the results at the baseline unit-level specification. Columns (2) and (3) report the equivalent destination–year shift-level IV regression, weighted by the exposure weights generated by `ssaggregate`. Column (2) clusters by destination country. Column (3) clusters by 16 geographic regions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Treatment dynamics — We next examine the treatment dynamics with four leads and six lags of DI^{depth}_{rjt} by estimating the following equation:

$$Y_{f,rjt} = \sum_{h=-4}^{6} \beta_h DI^{depth}_{rj,t-h} + \alpha_2 DI^{breadth}_{rjt} + \alpha_3 tariff_{rjt} + FE\ and\ Controls + \varepsilon_{f,rjt}. \tag{A2.1}$$

Figure A1 shows that none of the four leads is individually significant; they are also jointly insignificant ($F = 1.30$, $p = 0.267$). This alleviates concerns that innovation predicts subsequent deep integration of export destinations. The response at $h = 5$ is negative and statistically significant, whereas most of the other individual post-exposure coefficients are negative, although statistically insignificant.

The delayed response is consistent with Aghion et al. (2024), who find that patenting begins to respond approximately three years after a demand shock. Following their approach, we aggregate

America, North America, and South America; Central Asia, Eastern Asia, the Middle East, and Southern Asia; Central Europe, Eastern Europe, Northwestern Europe, Scandinavia, and Southern Europe; and Oceania.

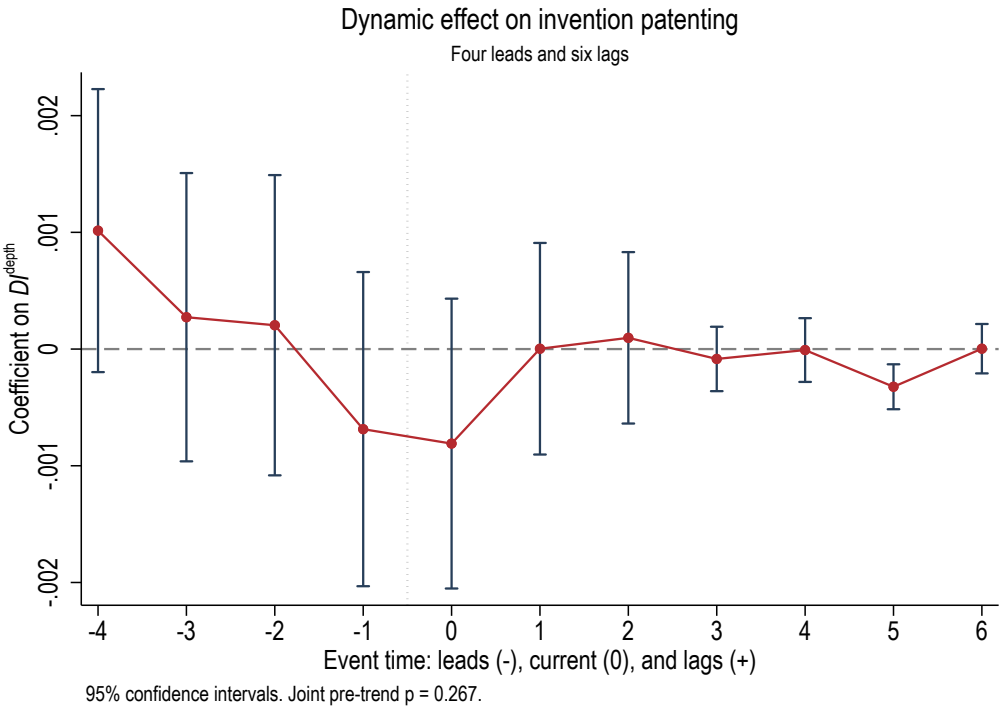


Figure A1: Dynamic effect of DI_{rjt}^{depth} on invention patenting. The figure reports the coefficients β_h from equation (A2.1) and 95% confidence intervals.

the dynamic effects into two windows: a contemporaneous and short-run window (years 0–1) and a medium-run window (years 2–6). Table A4 reports the corresponding linear combinations, with standard errors computed using the full covariance matrix. The cumulative effect from the contemporaneous period through lag 6 is -0.001124 ($p = 0.100$). A one-standard-deviation increase in DI^{depth} (3.957) therefore implies a reduction in the probability of patenting of approximately 0.445 percentage points. This corresponds to about 13.5% of the unconditional patenting probability.

Table A4: Cumulative dynamic response of patenting probability to depth exposure

Period	Event times	Coefficient sum	Standard error
Pre-trend	-4 to -1	0.000806	(0.000725)
Current	0 to $+1$	-0.000807	(0.000700)
Future	$+2$ to $+6$	-0.000317	(0.000402)
Post-exposure	0 to $+6$	-0.001124	(0.000683)

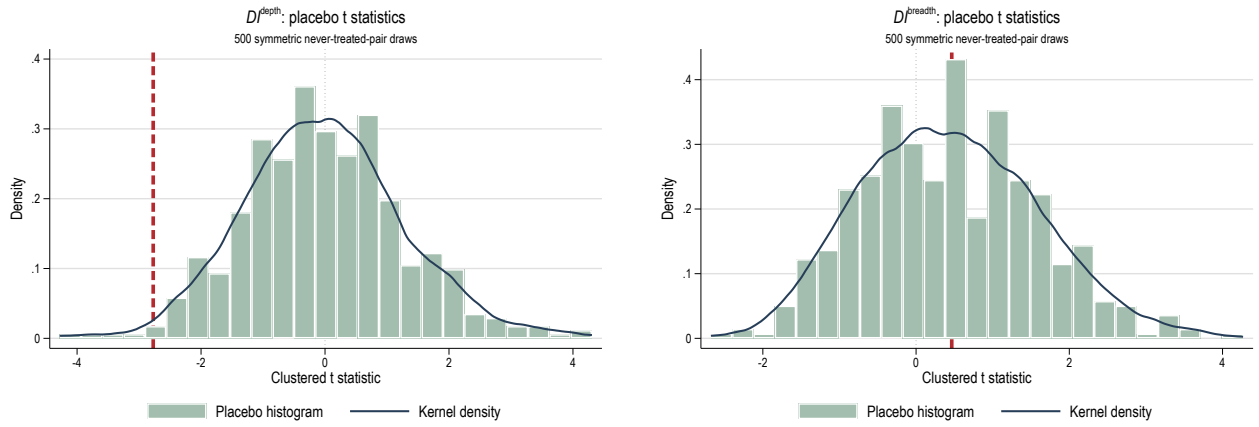
Notes: Each row reports a linear combination of coefficients from equation (A2.1). Standard errors use the full estimated variance-covariance matrix and therefore account for covariance among event-time coefficients. The corresponding p-values for the pre-trend, current, future, and post-exposure sums are 0.266, 0.249, 0.430, and 0.100. Standard errors are clustered at the city–industry level.

Permutation placebo test — To further address potential endogeneity concerns, we conduct a permutation-based placebo test. In each of 500 replications, we remove the observed PTA country pairs and randomly reassign their complete timing-and-depth histories to an equal number of never-treated country pairs, preserving the symmetry of bilateral PTA relationships. We then reconstruct both destination-level PTA engagement measures and the corresponding city–industry exposures, and re-estimate the baseline firm-level regression. Following Carr and Packham (2019), we plot the distribution of placebo t-statistics in Figure A2. The observed t-statistic for DI^{depth} is -2.772, with only 17 of the 500 placebo replications yielding a statistic at least as extreme in absolute value, corresponding to an empirical p-value of 0.034. In contrast, the observed t-statistic for $DI^{breadth}$ falls well within the distribution of placebo t-statistics. These results provide further support for our baseline finding, suggesting that the estimated negative effect of DI^{depth} is unlikely to arise from unobserved factors.

Table A5: Robustness checks: Alternative base-year exposure shares and recentered IV

	(1) Baseline	(2) 2005 shares	(3) 2010 shares	(4) Recentered IV
DI_{rjt}^{depth}	-0.000213***	-0.000142*	-0.000215**	-0.000195**
	(0.000077)	(0.000082)	(0.000090)	(0.000082)
$DI_{rjt}^{breadth}$	0.000017	0.000077**	0.000041	0.000073
	(0.000036)	(0.000036)	(0.000039)	(0.000056)
Tariff	-0.012173*	-0.002284***	-0.003036***	-0.011308
	(0.006865)	(0.000617)	(0.000611)	(0.006902)
Estimator	OLS	OLS	OLS	IV
Baseline *FE and Controls*	✓	✓	✓	✓
Kleibergen–Paap rk Wald F				20,201.91
Observations	2,541,619	2,738,309	2,815,212	2,541,619
Adjusted R^2	0.326	0.319	0.317	

Notes: The dependent variable is an indicator for positive invention patent filings. Standard errors clustered at the city–industry level are in parentheses. Columns (2) and (3) reconstruct DI^{depth} and $DI^{breadth}$ using shares with alternative base years: 2005 and 2010, respectively. Column (4) instruments the two realized DI measures using actual exposure minus the exposure expected under 500 symmetric permutations of complete timing-and-depth histories over the fixed set of ever-treated country pairs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.



(a) DI^{depth} (b) $DI^{breadth}$

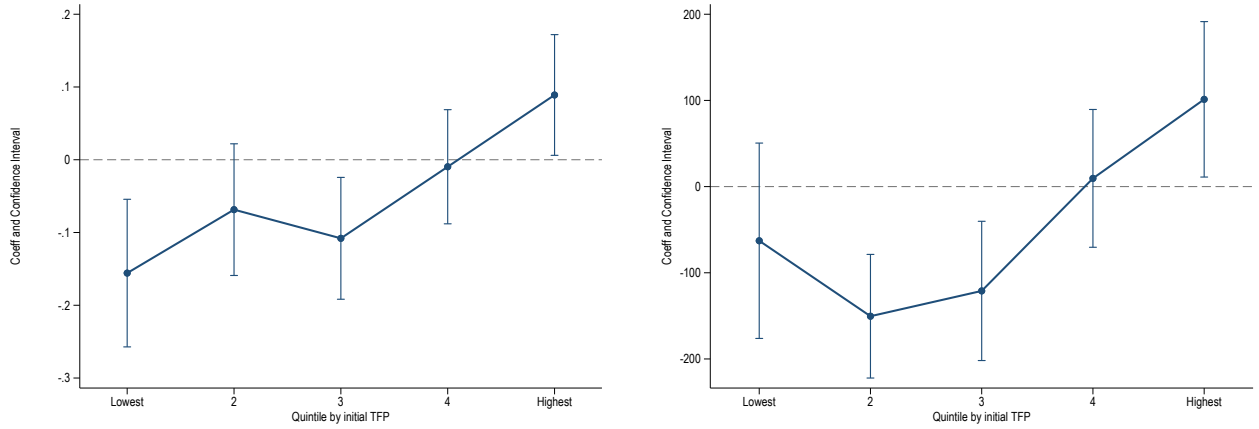
Figure A2: Permutation placebo based on 500 symmetric assignments to country pairs that never sign PTAs. Each panel plots the distribution of placebo t-statistics and its kernel density. The red dashed line is the observed baseline t statistic and the gray dotted line is zero.

Alternative base-year exposure shares — Table A5 reports sensitivity to the vintage of the

export shares. Column (1) uses the baseline 2000 shares, while Columns (2) and (3) reconstruct both exposure measures using 2005 and 2010 shares, respectively. The results indicate that the negative innovation effect of DI_{rjt}^{depth} is robust to alternative base-year exposure shares.

Recentered IV — Column (4) of Table A5 further addresses a potential identification concern arising from nonrandom exposure to PTA formation. Countries occupying different positions in the observed PTA network may have systematically different expected depth and breadth exposures. Following the recentering approach of Borusyak and Hull (2023), we treat the realized depth and breadth exposures as the endogenous regressors and instrument them with their deviations from the exposure expected under a specified PTA-assignment process. Specifically, we hold fixed the observed set of 3,110 country pairs that ever formed a PTA and randomly reassign the complete 1958–2015 agreement histories across these pairs. Each reassignment preserves the full timing-and-depth path of an agreement and assigns the same path in both directions of a country pair. For each of 500 draws, we reconstruct every destination country's average PTA depth and number of PTA partners and then aggregate them using the original city–industry destination shares. Averaging the resulting exposures across draws yields the depth and breadth exposures expected under the specified assignment process. Subtracting these expected exposures from their realized counterparts produces the two recentered instruments. The recentered-IV coefficient on DI^{depth} is statistically significant and close to the baseline estimate, while the coefficient on $DI^{breadth}$ remains statistically insignificant. This indicates that the negative depth effect is not driven by the component of DI exposure systematically predicted by the observed PTA network.

A3 Balanced-sample analysis and firm entry-exit dynamics



(a) Patenting probability

(b) Profit

Figure A3: Balanced-sample heterogeneous effects on patenting probability and profit by TFP. Each panel adopts the specification as in regression (5). Panel (a) shows the coefficient (percentage points) α_1^g for patenting probability, and panel (b) shows the corresponding coefficient for profit, for each TFP quintile g with 95% confidence intervals. The sample is restricted to firms observed in every year of the sample period. The dependent variable is an indicator for whether firm f files any invention patent in year t in panel (a). The dependent variable is firm profit in panel (b).

Table A6: Destination integration and firm entry-exit margins

	(1)	(2)	(3)	(4)
	Firm entry	First firm entry	Firm exit	First firm exit
DI_{rjt}^{depth}	0.000030	0.000111	0.000339**	0.000357**
	(0.000243)	(0.000115)	(0.000170)	(0.000170)
$DI_{rjt}^{breadth}$	0.000159**	-0.000017	-0.000012	-0.000035
	(0.000077)	(0.000058)	(0.000064)	(0.000065)
N	2,453,276	483,175	2,339,237	2,191,127

Notes: Standard errors clustered at the city–industry level are in parentheses. All regressions include the baseline controls. Firm entry equals one when a firm is observed in year t but not in $t-1$, equals zero when it is observed in both years, and is missing in the first sample year; re-entry after a gap is included. First firm entry equals one only for the first observed entry event, equals zero before that event, and is missing thereafter. Firm exit equals one when a firm is observed in year t but not in $t+1$, equals zero when it is observed in both years, and is missing in the last sample year; temporary departures are included. First firm exit equals one only for the first observed exit event, equals zero before that event, and is missing thereafter. Because inclusion in the industrial survey is subject to a size threshold, these variables measure observed sample entry and exit rather than necessarily firm birth and death. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A4 Other dimensions

Upstreamness — We examine heterogeneity along the dimension of firms' industry upstreamness. We measure upstreamness using the industry-level index proposed by Antràs et al. (2012), which is based on input–output tables and captures an industry's average distance from final use. We split industries into upstream and downstream groups using the median value of this index.

Figure A4 shows that both groups exhibit productivity heterogeneity. Compared with downstream industries, the adverse innovation response among the least productive firms tends to be stronger, whereas the positive innovation response among the most productive firms tends to be weaker in upstream industries. One possible explanation is that upstream industries, which are more closely involved in intermediate-input production, are particularly exposed to trade diversion induced by PTAs among export destinations. Consistent with this mechanism, Conconi et al. (2018) show that rules of origin in PTAs such as NAFTA substantially reduced imports of intermediate inputs from third countries relative to member countries. Such trade diversion may further discourage innovation among less productive Chinese firms in upstream industries and weaken the escape-competition

incentives of more productive firms.

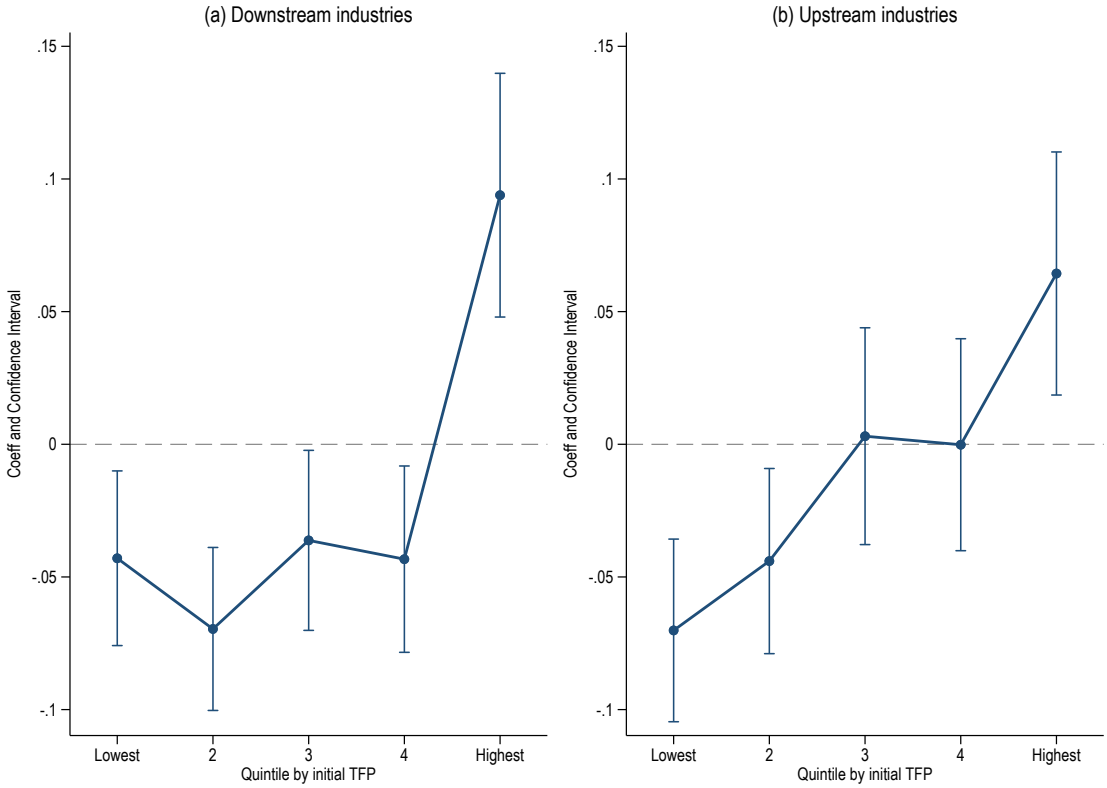


Figure A4: TFP heterogeneity in downstream and upstream industries. Panel (a) reports downstream industries and Panel (b) reports upstream industries. Each panel adopts the specification as in regression (5) and shows the coefficient (percentage points) α_1^g for each TFP quintile g and the corresponding 95% confidence interval. The dependent variable is an indicator for whether firm f files any invention patent in year t.

Local IPR enforcement — We further examine heterogeneity along the dimension of firms' local intellectual property rights (IPR) enforcement. Local IPR institutions shape firms' incentives to innovate by affecting the appropriability of innovation returns and the capability of escaping competition. We measure provincial IPR enforcement using the Intellectual Property Protection sub-index from the Marketization Index Report of Chinese Provinces compiled by Wang et al. (2016). This index is constructed from provincial indicators such as patent application intensity, efficiency in handling patent infringement cases, and the strength of judicial enforcement. Higher values indicate stronger IPR enforcement. Provinces are classified into low- and high-IPR groups based on the median

of the index.

Figure A5 shows that firms located in provinces with weaker IPR enforcement exhibit no statistically significant responses to PTA depth exposure across all five TFP quintiles. In contrast, firms located in provinces with stronger IPR enforcement exhibit pronounced heterogeneous effects across productivity quintiles. The most productive firms increase their patenting propensity in response to PTA depth exposure, whereas the least productive firms exhibit a decline in patenting propensity.

These findings suggest that the innovation effects of deep integration among export destinations depend on local institutional conditions. Stronger IPR enforcement enhances more productive firms' ability to appropriate the returns to innovation, thereby strengthening their incentives to innovate and escape competition. By contrast, less productive firms may rely more heavily on imitation and technology adoption, which become more costly under stronger IPR protection. Consequently, these firms may have greater difficulty in responding to competition through innovation, leading to a decline in patenting propensity. Overall, deep integration among export destinations is more likely to stimulate innovation among highly productive firms when supported by strong local IPR institutions.

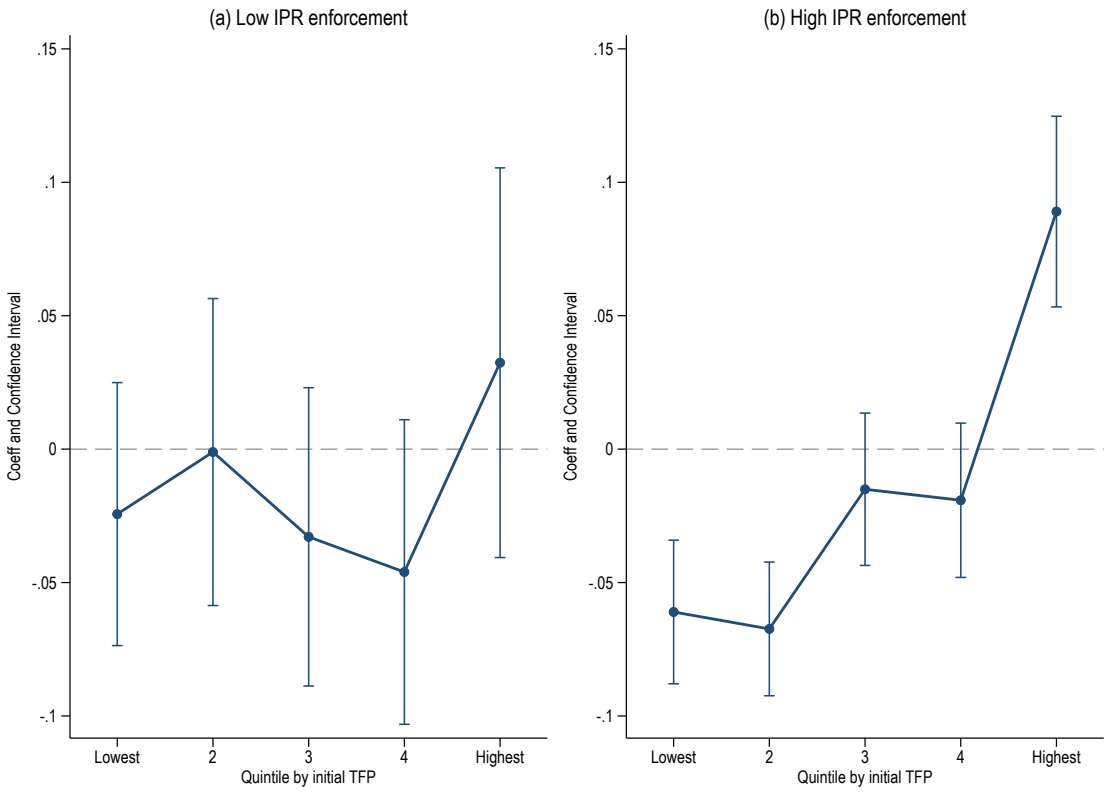


Figure A5: TFP heterogeneity in low and high local IPR enforcement. Panels (a) and (b) report provinces with low and high IPR enforcement, respectively. Each panel adopts the specification as in regression (5) and shows the coefficient (percentage points) α_1^g for each TFP quintile g and the corresponding 95% confidence interval. The dependent variable is an indicator for whether firm f files any invention patent in year t.

Ownership — Finally, we explore heterogeneous innovation responses across firms with different ownership types: state-owned, private, and foreign-owned. Table A7 shows that all three types of firms exhibit negative responses in patenting propensity to PTA depth exposure, but only the estimate for private firms is statistically significant.

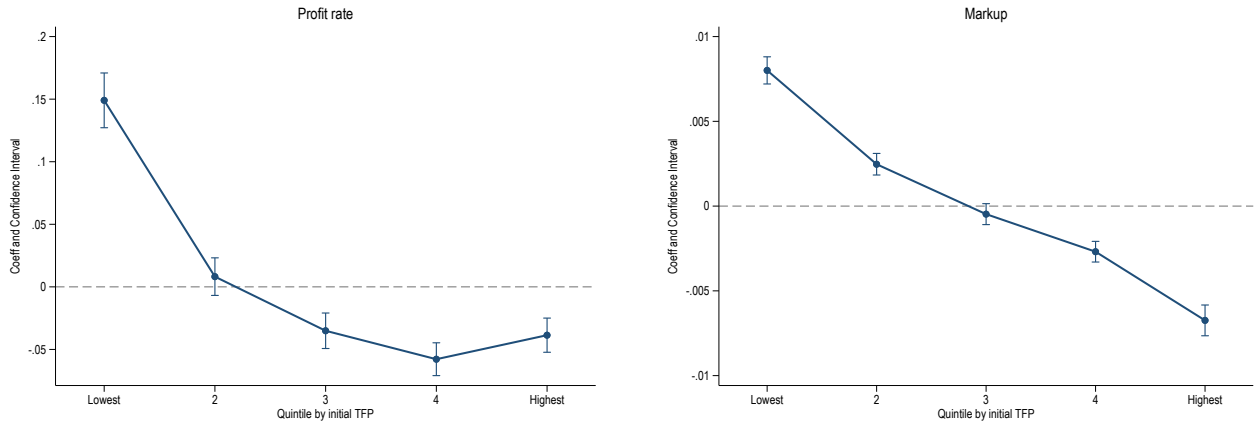
One possible interpretation is that state-owned firms are partially insulated from short-run competitive pressure due to their soft budget constraints and government support. Foreign-owned firms may be able to reallocate innovative activities across affiliates, thereby mitigating the impact of competitive pressure. These results should nevertheless be interpreted cautiously because the sample sizes for state-owned and foreign-owned firms are relatively small compared with that for private firms.

Table A7: Destination integration and patenting by ownership

	(1)	(2)	(3)
	State-owned	Private	Foreign
DI_{rjt}^{depth}	-0.000188	-0.000276***	-0.000356
	(0.000269)	(0.000081)	(0.000720)
$DI_{rjt}^{breadth}$	0.000270	-0.000013	0.000171
	(0.000192)	(0.000036)	(0.000157)
Observations	180,913	2,212,035	146,965

Notes: We split the sample based on firms' ownership and adopt the baseline specification for each ownership subsample. The dependent variable is an indicator for whether a firm files at least one invention patent in year t. Standard errors clustered at the city–industry level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A5 Heterogeneous effects on profit rate, markup, and export scope



(a) Profit rate

(b) Markup

Figure A6: TFP heterogeneity in firm-level profit rate and markup. Each panel adopts the specification as in regression (5) and shows the coefficient α_1^g for each TFP quintile g and the corresponding 95% confidence interval. The dependent variable is firm-level profit rate in panel (a) and markup in panel (b), respectively. Both measures are winsorized at the 1st and 99th percentiles.

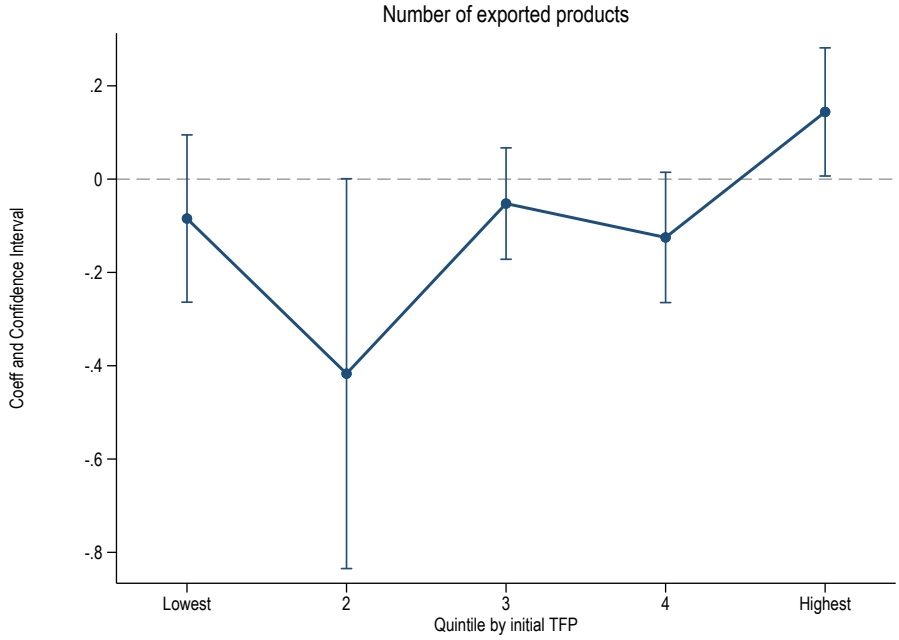


Figure A7: TFP heterogeneity in firm-level export scope. The panel adopts the specification as in regression (5) and shows the coefficient α_1^g for each TFP quintile g and the corresponding 90% confidence interval. The dependent variable is the number of products exported by each firm in a given year.

A6 Additional tables

Table A8: Poisson regressions for invention patent counts

	(1)	(2)	(3)
DI_{rjt}^{depth}	-0.011553	0.006657	0.004946
	(0.028892)	(0.008023)	(0.007539)
$DI_{rjt}^{breadth}$	-0.004862	0.000316	0.000378
	(0.004944)	(0.002176)	(0.002226)
Tariff	-4.843933*	0.892684	1.017366
	(2.700942)	(0.652641)	(0.629040)
Firm FE	No	Yes	Yes
Firm characteristics × year	No	No	Yes
City-year FE	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes
City-Industry FE	Yes	Yes	Yes
Market size exposure	Yes	Yes	Yes
Observations	2,100,206	299,947	299,947
Pseudo R^2	0.502	0.810	0.812

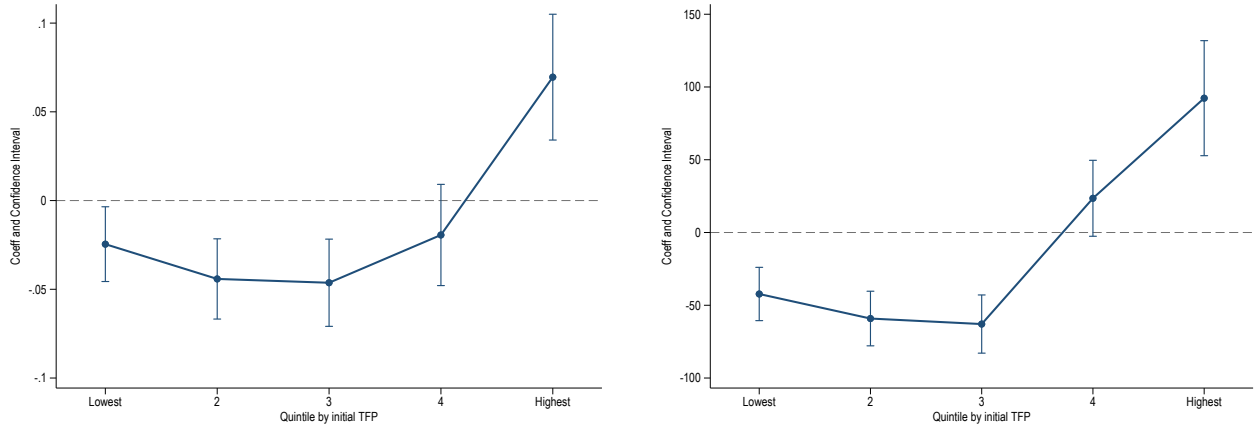
Notes: The dependent variable is the invention patent filing count, including zeros. Coefficients are PPML semi-elasticities. Standard errors clustered at the initial city–industry level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Categorization of WTO+ and WTO-X provisions

WTO+	WTO-X	
Tariff industrial goods	***Competition policy***	Health
Tariff agricultural goods	***Investment measures***	Human rights
Customs	***Movement of capital***	Illegal immigration
Export taxes	***Intellectual property rights (IPR)***	Illicit drugs
SPS measures	Labor market regulation	Industrial cooperation
State trading enterprises	Environmental laws	Information society
TBT measures	Visa and asylum	Anti-corruption
Countervailing measures	Consumer protection	Mining
Anti-dumping	Data protection	Money laundering
State aid	Approximation of legislation	Nuclear safety
Public procurement	Agriculture	Political dialogue
TRIMS measures	Audiovisual	Public administration
GATS	Civil protection	Regional cooperation
TRIPS	Innovation policies	Research and technology
	Cultural cooperation	SMEs
	Economic policy dialogue	Social matters
	Education and training	Statistics
	Energy	Taxation
	Financial assistance	Terrorism

Note: The provisions in bold are the core provisions, of which the MFN provisions are italicized. Non-italicized bold provisions are preferential ones.

A7 Additional figures



(a) Patenting probability

(b) Profit

Figure A8: Robustness: heterogeneous effects on patenting probability and profit using ACF TFP. Panel (a) shows the coefficient (percentage points) α_1^g of regression (5) for patenting probability, and panel (b) shows the corresponding coefficient for profit, for each initial within-industry ACF TFP quintile g with 95% confidence intervals. Panel (a) uses an indicator for whether firm f files any invention patent in year t; panel (b) uses firm profit.

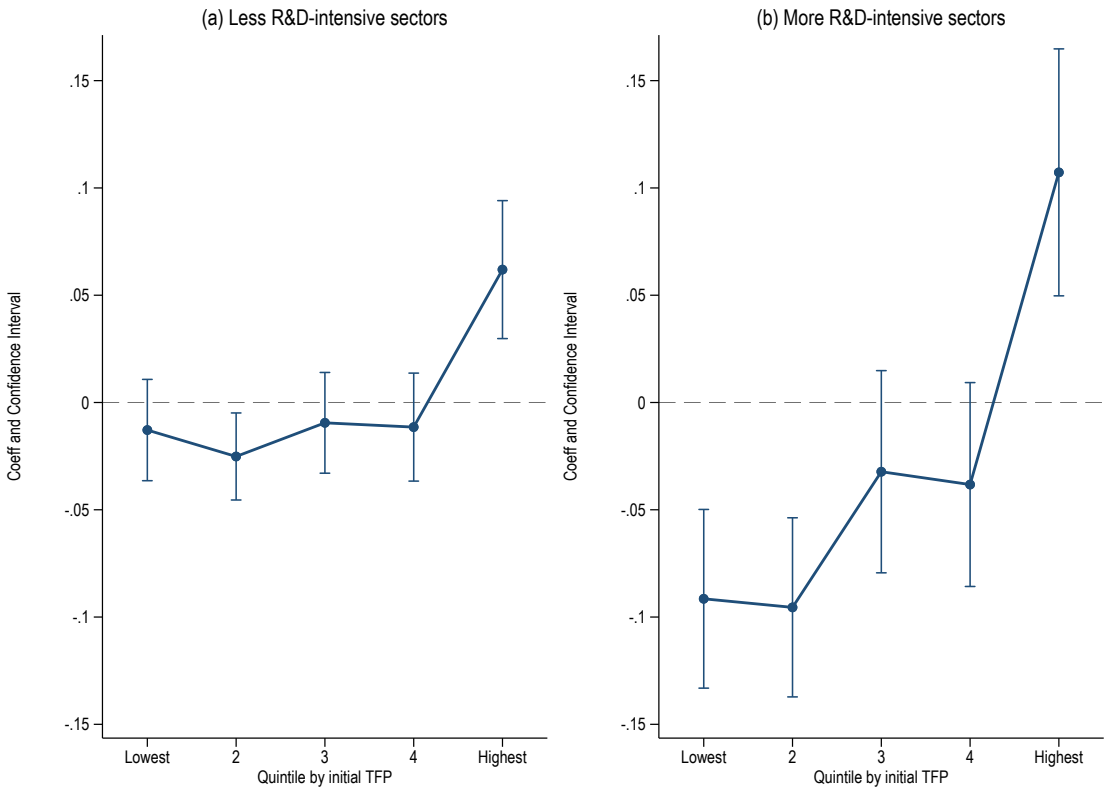


Figure A9: Heterogeneous innovation effects by TFP and R&D intensity. The plots show the coefficient (percentage points) α_1^g of regression (5) for each TFP quintile g and the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f's invention patenting probability. The left plot focuses on the subsample of sectors with low R&D intensity, and the right plot focuses on the subsample of sectors with high R&D intensity.

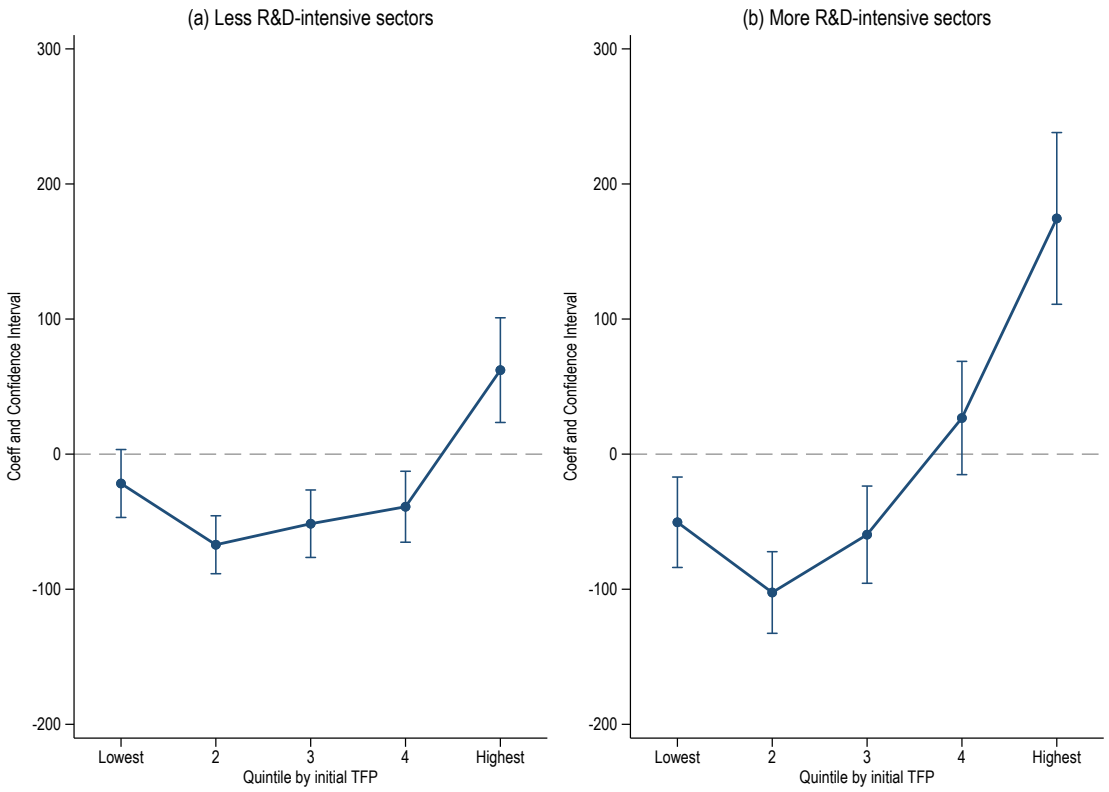


Figure A10: Heterogeneous profit effects by TFP and R&D intensity. The plots show the coefficient α_1^g of regression (5) for each TFP quintile g and the corresponding 95% confidence interval, where g ranges from 1 to 5. The dependent variable is firm f's profit level. The left plot focuses on the subsample of sectors with low R&D intensity, and the right plot focuses on the subsample of sectors with high R&D intensity.